



An explainable deep learning approach for automated detection and grading of diabetic retinopathy from fundus images

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ABSTRACT

Diabetic Retinopathy (DR) is a leading cause of preventable blindness, and it is important to accurately detect it as early as possible, as well as to measure the severity to provide early clinical indicators. Fundus image assessment in the manual mode is subjective and labor-intensive, and is hard to scale, which has encouraged automated approaches. Current deep learning methods are usually based on either convolutional neural networks (CNNs) or transformer-based ones, focusing on local lesion features or global retina context separately. In addition, relational dependencies among lesions that are clinically significant in the case of severity development are under-modeled. This paper introduces a relational, hybrid deep learning model that integrates convolutional, transformer-based, and graph attention models to classify binary and multiclass DR. A ResNet-Graph Attention Network (ResNet-GAT) is introduced to explicitly model spatial relational dependencies among regional feature descriptors and is compared with independent stand-alone ResNet-50 and Swin Transformer models. Also, a hybrid ResNet-Swin Transformer model combines fine-grained lesion representations and hierarchical global context. Experiments with the APTOS 2019 data show that the hybrid model has an accuracy of 98.09% with a Quadratic Weighted Kappa (QWK) of 0.9618 when used to classify binary, and 94.89% with a QWK of 0.9726 when used to classify five classes of severity. Robustness is proven by cross-dataset inference on IDRiD. The interpretability analysis performed through Grad-CAM shows that the predictions are made based on clinically significant areas, such as microaneurysms, hemorrhages, and exudates. It is demonstrated that performance and interpretability are enhanced by a combination of local, global, and relational representations in automated screening of DR.

1. Introduction

DR is a microvascular complication of diabetes mellitus and is one of the major causes of avoidable vision loss and blindness in the world. The reports of global health indicate that over 530 million individuals are now living with diabetes, and around 30–35% of them acquire at least some form of diabetic retinopathy at some point in their lives (Dai et al., 2021). Out of those who are affected, it advances to vision-threatening levels like Severe or Proliferative DR in close to 10% (Oh et al., 2021). This is a progressive disease that occurs with lesions on the retina, such as microaneurysms, hemorrhages, hard exudates, and

neovascularization, and thus the need to detect the disease at the initial stages and ensure proper severity levels to allow clinical intervention (Chen et al., 2021). Fig. 1 shows the key pathological features in a retinal fundus image that are affected by diabetic retinopathy.

The most popular modality that has been adopted in the screening of DR is the retinal fundus imaging because it is a non-invasive system and low in cost (Chetoui and Akhloufi, 2020). The traditional diagnosis is done by manual inspection through fundus images by ophthalmologists, whereby the severity of the disease is determined by the presence and distribution of lesions. As much as this method is clinically acceptable, it is labor-intensive, prone to inter-observer differences, and hard to scale

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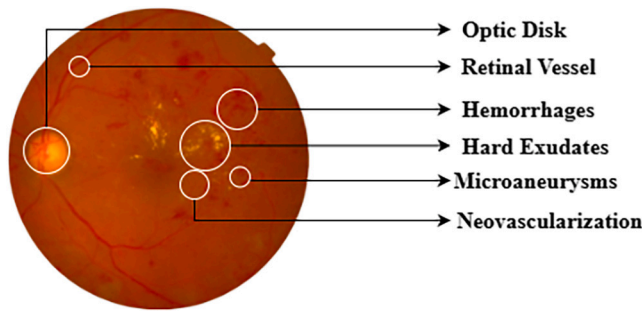


Fig. 1. Key pathological features in a retinal fundus image affected by diabetic retinopathy (microaneurysms, hemorrhages, neovascularization, and hard exudates).

up to screening of the population, particularly in low-resource contexts (Ayala et al., 2021). These limitations have prompted the use of machine learning-based solutions based on automated machine learning in the analysis of DR.

Early machine learning methods of DR detection used handcrafted features like color statistics, texture descriptors, and morphological features, and coupled them with classical classifiers like support vector machines and random forests (Gadekallu et al., 2020). These techniques showed decent performance in controlled settings, but sensitivity to feature design, change in image quality, and bias in the data sets made them very sensitive and restricted their strength and generalizability in clinical environments (Nguyen et al., 2020).

There are still several problems with automated DR detection. Most of them note that there are constraints like unbalanced datasets, differences in the quality of the images, and a lack of generalization between multiclass severity grading. There are some interpretability tools that have been applied, including Grad-CAM, which have been employed to point out the clinically relevant lesion areas, thus improving transparency in clinical settings.

Deep learning became a significant boost to automated DR analysis as it allowed end-to-end learning of features through the raw fundus images. Convolutional neural networks (CNNs) and especially deep-based models like ResNet-50 have demonstrated a high ability to identify fine-grained lesion-level content and have obtained high accuracy in binary and multiclass-based DR classification problems (Mishra et al., 2020; Jabbar et al., 2024). Nonetheless, CNNs are inherently local receptive fields and in many cases fail to learn long-range dependencies and global retinal structure, which are relevant to measuring advanced stages of DR with extensive pathological changes (Chaturvedi et al., 2020).

In order to overcome such shortcomings, more recently, transformer-based architectures have been applied to the study of retinal images. Other recent models like the Swin Transformer use hierarchical self-attention to achieve global contextual information and multi-scale dependencies across the retina, resulting in better severity grading performance in several publications (Pandey and Alsheikheh, 2025; Romero-Oraá et al., 2024). Simultaneously, graph-based deep learning frameworks, such as Graph Attention Networks (GATs), have been investigated to represent relational dependencies among salient regions or lesions, allowing clear expression of lesion co-occurrence and spatial interactions that are clinically related to disease progression (Rachapudi et al., 2023; Wahab Sait, 2023).

Although these developments have taken place, there are several weaknesses in the existing studies. Methods based on CNN give priority to local lesion characteristics but do not have a sense of context at a global scale, whereas transformer-based methods can miss minor lesion-level features that contribute to the early-stage detection of DR (Chaturvedi et al., 2020; Pandey and Alsheikheh, 2025; Romero-Oraá et al., 2024). Graph-based models, despite having a proven ability to do relational reasoning, are not as effective when applied in isolation,

and can fail to give a full picture of the retinal context (Rachapudi et al., 2023). In addition, the majority of the previous literature concentrates on a single architectural paradigm and fails to approach the two complementary advantages of convolutional, transformer-based, and graph-based representations in a coherent framework (Mohanty et al., 2023). Consequently, the architectural trade-offs affecting performance at various levels of severity of DR have not yet been studied thoroughly.

This experiment mitigates these shortcomings in the given work and suggests a single deep learning architecture that incorporates convolutional, transformer-based, and graph-based models to classify diabetic retinopathy. In particular, this study uses ResNet-50 to learn fine-grained lesion-level features, Swin Transformer to learn hierarchical global retinal context, and a ResNet-Graph Attention Network (ResNet-GAT) to explicitly learn inter-lesion relational dependencies. On these elements, this study then suggests a hybrid ResNet-50 + Swin Transformer model and designs a model that integrates local lesion sensitivity with global contextual awareness. The framework is tested on the APTOS 2019 dataset, binary and five-class DR grading tasks, and further inferences on the IDRid dataset to determine if the framework is robust to datasets.

The key contributions of this paper are summarized here:

1. Unified DR classification modeling paradigms: This study evaluates ResNet-50 (CNN), Swin Transformer, and a ResNet-Graph Attention Network (ResNet-GAT) to address the role of local feature extraction, global contextual modeling, and inter-lesion relational reasoning in binary and multiclass diabetic retinopathy recognition.
2. Relational lesion modeling via ResNet-GAT: This experiment suggests a ResNet-GAT framework that models spatial relations among regional feature descriptor connections in fundus retina images and proves that relational reasoning enhances the dissimilarity of clinically tricky stages of DR, especially the Mild and Moderate classes.
3. Interpretable framework: This experiment developed a hybrid CNN-Transformer model that combines lesion-level and global retinal representations and proves to be more accurate and with better Quadratic Weighted Kappa at the same time, with Grad-CAM-based explainability that enables clinical interpretability.

The rest of the paper is structured in the following way. In Section 2, recent work in related areas related to automated diabetic retinopathy detection is reviewed. Section 3 presents the suggested methodologies and model architectures. Section 4 describes datasets, preprocessing, and experimental setup. The results and discussions are presented in Section 5. Section 6 concludes the paper with directions on future work.

2. Related works

The recent progress in automated detection of diabetic retinopathy (DR) has been dominated by methods of deep learning on retinal fundus images. In the last few years, most studies can be categorized into convolutional neural network-based methods, transformer-based methods, graph-based relational methods, and hybrid approaches, which can involve many paradigms at once.

Convolutional neural networks (CNNs) are being widely used in detecting and grading the severity of DRs as they have a high ability to learn hierarchical features of images. Several studies used deep CNN models, including ResNet, DenseNet, and EfficientNet, to do binary and multiclass DR classification. As an example, Mishra and Hanchate proposed a CNN-based model on the APTOS 2019 dataset and presented over 96% accuracy on DR detection (Mishra et al., 2020). In a similar case, Bodapati et al. (2021) proposed a deep convolutional feature aggregation framework that predicts DR severity and demonstrated improved performance across multiple severity levels (Bodapati et al., 2021). On the same note, Ayala et al. applied the use of transfer learning using DenseNet-based architectures and obtained high classification accuracy in various datasets (Ayala et al., 2021). In a related

study, Mostafa et al. (2025) suggested a CNN-based classification model that effectively graded DR severity stages using fundus images (Mostafa et al., 2025). Although the performance of these methods was good in the context of detecting the lesion-level patterns, they were constrained by local receptive fields, hence were weak in detecting global retinal structure, especially in later stages of DR when pathological changes were widespread.

Transformer-based architectures have most recently been introduced into retinal image analysis to provide better contextual modeling over the whole world. Romero-Oraa and Herrero-Tudela (a) suggested an attention-dependent model, which independently manipulated bright and dark retinal structures to enhance the level of DR grading, and reported higher classification performance on benchmark datasets (Romero-Oraa et al., 2024). In a similar study in the previous year (2023), Khaparde et al. developed an attention-based Swin U-Net framework integrated with hybrid deep learning models, where retinal lesions were segmented and classified using deep neural networks, which gave an improvement in classification results. However, there was architectural complexity and the framework relied on multiple optimization and segmentation stages, which increased computational cost and limited scalability (Khaparde et al., 2023). Pandey and Alsheikheh used deep CNNs with attention mechanisms and achieved better severity grading in the results (Pandey and Alsheikheh, 2025). In the same year (2025), Gaddam proposed an attention-guided and sequence-aware classification for the DR framework, achieving very high accuracy across multiple datasets. But the proposed pipeline increased architectural complexity and computational cost (Gaddam, 2026).

In another approach, hierarchical transformer-based models have been investigated, including the Swin Transformer to learn long-range dependencies as well as multi-scale retinal representations to achieve better Quadratic Weighted Kappa scores in multiclass DR grading problems (Oh et al., 2021). In the following years, not just a hierarchical, but also a self-supervised model was proposed. For example, Ohri et al. (2024) suggested a self-supervised vision transformer-based study to detect and classify diabetic retinopathy, which improved Quadratic Weighted Kappa scores by leveraging large-scale unlabeled fundus images. However, the proposed framework required a lot of pretraining data and huge computational resources, limiting its practicality in low-resource settings (Ohri et al., 2024). Nevertheless, transformer-based models can be characterized by a large training set that is necessary, and are less sensitive to small, localized lesions that are important in early-stage DR detection.

Deep learning graph models are a new trend in retinal image modeling of relational dependence. Graph Neural Networks (GNNs) allow the lesion interactions to be explicitly modeled by the encoding of salient regions or lesion candidates as nodes and the establishment of edges between the nodes through the interaction of spatial or semantic relationships. Rachapudi and Rao used an optimized deep learning model that has integrated relational reasoning and was reported to be very accurate on benchmark DR datasets (Rachapudi et al., 2023). In 2023, Dhinakaran et al. introduced a semi-supervised graph learning framework to detect diabetic retinopathy that boosted relationships between labeled and unlabeled retinal images, which improved grading performance on imbalanced datasets. Although the method relied on handcrafted graph construction and classical feature preprocessing, limiting its integration with end-to-end deep representation learning (Dhinakaran et al., 2023). Wahab Sait suggested the lightweight DR detection framework, which is based on graph-based feature selection and has competitive performance on the APTOS and EyePACS data sets (Wahab Sait, 2023). Although these methods proved the significance of relational modeling, the choice of graph construction strategy often limited them, and they were usually tested separately, without consideration of the global context of information.

Dubious deep learning techniques are also suggested to enable the two modeling paradigms to realize their complementary advantages.

Jabbar and Liaqat proposed a lesion-oriented hybrid deep learning, which incorporated CNN feature extraction methods alongside optimization techniques, which was more accurate than single CNNs (Jabbar et al., 2024). Mohanty et al. have compared various deep learning structures and reported that DenseNet-based models are better than various hybrid structures in multiclass DR classification (Mohanty et al., 2023). Even though hybrid CNN-based methods have demonstrated better robustness, most current hybrid methods pay little attention to the contribution of relational lesion modeling in addition to global contextual representations, concentrating on feature-level fusion only.

In general, the existing literature shows that CNN-based models are useful to capture fine-grained lesion features, transformer-based structures enhance the global contextual perception, and the use of graphs allows the researcher to perform relationship reasoning through retinal structures. Nevertheless, they are mostly studied in isolation or in a few combinations, and there is still a lack of systematic studies that simultaneously investigate the features of local lesions, the global retinal context, and the inter-relationships among lesions within a single experimental design.

3. Materials & method

The following section illustrates the suggested approach to automated diabetic retinopathy (DR) classification. The three clinical and computational observations that drive the design of the framework are: (i) small, localized lesions of the early stages of DR involve fine spatial sensitivity, (ii) more advanced stages of DR require extensive structural changes, requiring global contextual knowledge, and (iii) The spatial and relational patterns of multiple lesions affect disease progression and not individual findings.

To cover the aspects in a unified manner, the suggested methodology incorporates convolutional, transformer-based, and graph-based modeling paradigms. The section is structured with a bottom-up approach, beginning with data flow and moving on to individual architecture elements, fusion strategy, training goals, and interpretability.

3.1. Proposed framework overview

The overall structure is that of a deterministic and reproducible data flow represented as in Fig. 2, in which a retinal fundus image is converted into a DR prediction by a series of clearly defined stages. The architecture is made in such a way that dependency cascades will not choke off feature diversity; hence, parallel representation learning is embraced, as opposed to sequential feature passing.

The general process flow of the proposed approach is shown in Fig. 2.

This feature fusion ensures that neither local lesions nor global retinal contextual structures dominate the learning process for the model.

3.2. Local lesion-level feature extraction using ResNet-50

The convolutional backbone selected is ResNet-50 because it is an effective model in the analysis of medical images, and it has the capability of retaining low-level spatial features via residual learning. The properties are significant in the context of DR when it is necessary to determine small pathological structures, including microaneurysms and dot hemorrhages.

Every residual block is trained on a residual function instead of a mapping. Considering the input feature map x , the output of a residual block is as follows:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (1)$$

The ResNet-50 architecture constitutes 50 layers, sequenced as follows: starts with a convolutional layer with 7×7 filters, trailed by 16 residual blocks arranged into four stages. A bottleneck design to improve computational efficiency is employed in each residual block. A global

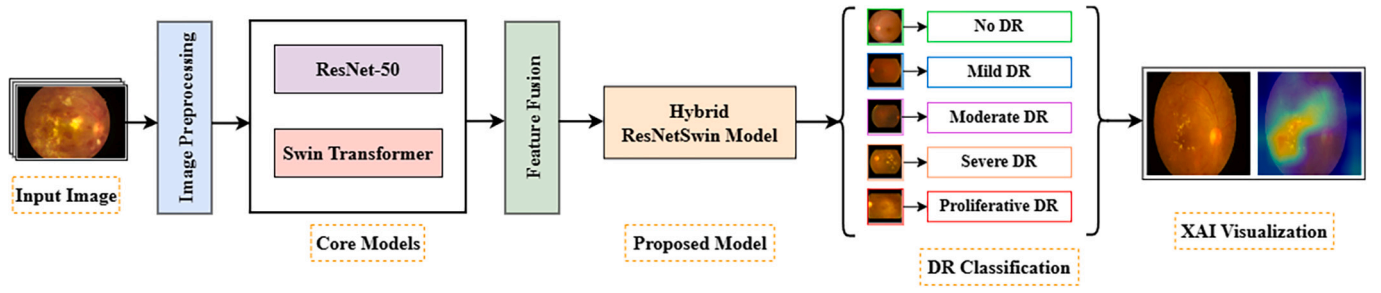


Fig. 2. Proposed Methodology Flow Diagram. Local lesion features and global retinal context are learned via CNN-based and Transformer-based models and fused for the final model to classify DR.

average pooling layer is applied before the final fully connected layer after the residual blocks.

Kernels W are used in each convolutional operation employed to extract features such as local textures, vessel boundaries, and lesion information. If W represents the convolutional kernels, x is the input feature map, and b is the bias term, then the convolutional process can be mathematically expressed as:

$$y = W * x + b \quad (2)$$

Kernels W are applied in each convolutional operation to extract local texture, vessel boundaries, and lesion information, computed as:

$$f_{i,j,k} = \sigma \left(\sum_{m,n} W_{m,n,k} \cdot x_{i+m,j+n} + b_k \right) \quad (3)$$

where $f_{i,j,k}$ represents the feature activation at a spatial location, (i, j) and k, b_k are the biases and σ denoted as the activation function (ReLU). Such a formulation would enable the network to transmit information that is sensitive to lesions across deep layers without damage. Following the last convolutional step, global average pooling is used to reduce spatial data to a fixed-length representation:

$$f_{ResNet} = GAP(y) \quad (4)$$

ResNet-50 architecture productively extracts the low-level, fine-grained features such as hard exudates, microaneurysms, and vessel structures, which are essential to detect DR stages.

The ResNet-50 architecture diagram is shown in Fig. 3.

3.3. Global contextual feature extraction using Swin Transformer

To complement local feature extraction, a Swin Transformer branch is used to capture long-range dependencies and global structural patterns of the retina. Compared to traditional vision transformers, Swin Transformer has a hierarchical design with shifted window-based self-attention, which is computationally efficient with high-resolution medical images.

This separates the input image into non-overlapping patches, and then the embedded patches are linearly concatenated into a token sequence:

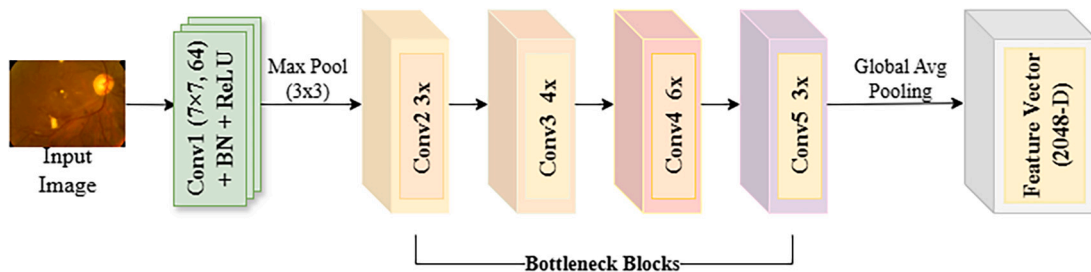


Fig. 3. ResNet-50 architecture to extract local lesion-level features.

$$X = [x_1, x_2, \dots, x_N], x_i \in \mathbb{R}^{P^2 \times C} \quad (5)$$

where, $P \times P$ represents the patch size and C denotes the number of channels.

In a window, the multi-head self-attention operation is computed as follows:

$$MHSA(Q, K, V) = \text{Contact}(\text{head}_1, \dots, \text{head}_h) W^0 \quad (6)$$

where the attention head is defined as:

$$\text{head}_i = \text{Softmax} \left(\frac{Q_i K_i^T}{\sqrt{d_k}} + B \right) V_i \quad (7)$$

The shifted window mechanism of Swin Transformer activates information exchange between neighboring windows across layers, which allows the model to gradually build a global representation of the retinal structure.

The Swin Transformer architecture diagram is shown in Fig. 4.

3.4. Relational lesion modeling: ResNet-GAT

Though CNNs and transformers implicitly describe spatial dependencies, they do not explicitly describe interrelationships between individual lesion regions. A ResNet-Graph Attention Network (ResNet-GAT) is proposed to model spatial relational patterns among regional feature descriptors, capturing the co-occurrence and spatial arrangement of lesion-bearing regions. The relational branch is located on top of the output from the last residual stage (layer4/conv5) of the ResNet-50 backbone network, which, for a 224×224 input, yields a feature map F of size $2048 \times 7 \times 7$ (2048 channels over a 7×7 spatial grid). Each of the 49 spatial positions on this map is considered a graph node, and its channel vector is a 2048-dimensional vector. Hence, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has 49 nodes and node-feature dimension 2048. A saliency selection/lesion detector is not employed; all spatial positions provide a node, and the graph-attention layers are trained to identify informative regions and relations.

The edges are formed as a result of spatial adjacency, which is 4-connected in the feature-map lattice; the nodes are connected if and only if

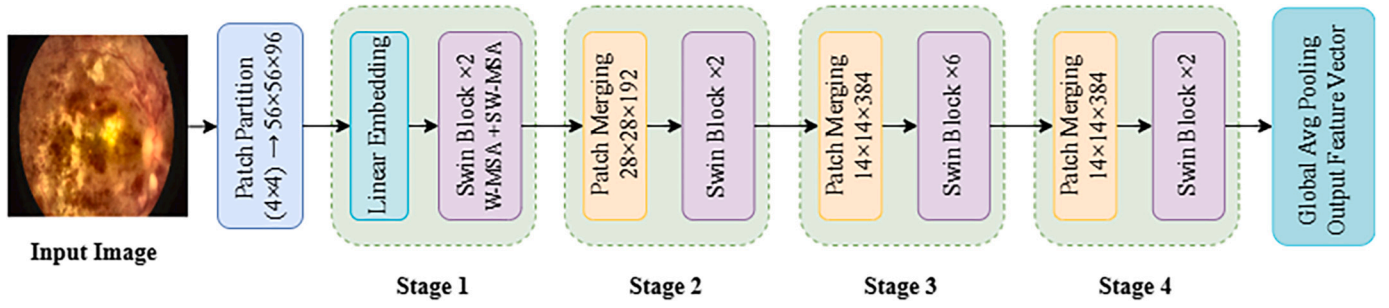


Fig. 4. Swin Transformer Architecture for capturing the global retinal contextual features.

$|i - \hat{i}| + |j - \hat{j}| = 1$. So, each interior node has exactly four adjacent nodes, and for the lattice of size 7×7 , there are $|\mathcal{E}| = 168$ directed edges (or 84 undirected edges). Since adjacency is topological and not distance/gated or similarity-gated, there is no threshold to learn or tune, and the same edge set is generated for all images, thus keeping the relational branch parameter-light and fully reproducible. Edge-to-edge message passing allows each region to build up its context from its spatial neighbourhood, which models the co-occurrence of retinal structures in that neighbourhood.

The attention-weighted aggregation is used to update node representations:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} a_{ij} W h_j \right) \quad (8)$$

With attention coefficients:

$$a_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i || W h_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [W h_i || W h_k]))} \quad (9)$$

Here, the ResNet-GAT graph construction is shown via an algorithm that covers ResNet feature extraction to classification through a softmax output.

Algorithm 1. ResNet-GAT graph construction and forward pass.

Input: fundus image ($3 \times 224 \times 224$)	
1: $F = \text{ResNet50_features}(x)$	# final stage; F in $R^{2048 \times 7 \times 7}$
2: $V = \text{flatten spatial cells of } F$	# 49 nodes, each v_i in R^{2048}
3: $E = \{(a, b): a, b \text{ are 4-neighbour grid-adjacent}\}$	# $ E = 168$ directed
4: $H = \text{ELU}(\text{GATConv}_1(V, E))$	# $2048 \rightarrow 256$, heads=4, concat $\rightarrow 1024$
5: $H = \text{Dropout}(H, p=0.3)$	
6: $H = \text{ELU}(\text{GATConv}_2(V, E))$	# $1024 \rightarrow 256$, heads=1 $\rightarrow 256$
7: $Z = \text{mean}_n H_n$	# global mean pooling $\rightarrow 256$
8: $y_{\text{hat}} = \text{softmax}(W_z + b)$	# linear classifier, C classes
Output: class probabilities y_{hat}	

The ResNet-GAT architecture configurations are shown in Table 1.

Table 2 shows the ResNet-GAT training configurations.

Fig. 5 shows the ResNet-GAT architecture that explains relational lesion modeling.

The architecture diagram explains the simplified overview of the ResNet-Graph Attention Network (ResNet-GAT). ResNet-50 extracts a

Table 1
ResNet-GAT architecture configuration.

Component	Setting
Backbone	ResNet-50 (ImageNet-pretrained), features_only, final stage (layer4)
Feature map	$2048 \times 7 \times 7$
Nodes $ V $	49 (all spatial locations); node dimension 2048
Edges $ E $	4-neighbour grid; 168 directed (84 undirected); unweighted; no threshold
GAT layer 1	GATConv $2048 \rightarrow 256$, heads = 4, concat $\rightarrow 1024$; ELU
Dropout	0.3
GAT layer 2	GATConv $1024 \rightarrow 256$, heads = 1 $\rightarrow 256$; ELU
Readout	global mean pooling over 49 nodes $\rightarrow 256$
Classifier	Linear $256 \rightarrow C$ ($C = 2$ binary, 5 multiclass)

Table 2

ResNet-GAT Training Configuration.

Hyper-parameter	Vale
Input size	224×224
Batch size	16
Optimizer	AdamW, weight decay $1e-4$ (1×10^{-4})
Learning Rate	$1e-4$ (1×10^{-4})
Epochs	up to 25 (best val-QWK checkpoint retained)
Loss	Focal loss ($\alpha = 1.0, \gamma = 2.0$)
Class balancing	WeightedRandomSampler (inverse class frequency)
Random seed	42 (Python, NumPy, PyTorch; PYTHONHASHSEED fixed)
Framework	PyTorch + PyTorch Geometric (GATConv) + timm

$7 \times 7 \times 2048$ feature map from its final residual stage, and the 49 spatial cells of this map serve as graph nodes (regional feature descriptors), connected by a fixed 4-neighbour spatial lattice.

3.5. Hybrid feature fusion and classification

The proposed final model consists of local and global representations that are fused at the feature level. ResNet-50 and Swin Transformer branches work in parallel and generate feature vectors f_{ResNet} and f_{Swin} .

After extracting lesion-focused local feature representations from ResNet-50 (f_{ResNet}) and global feature maps from Swin Transformer (f_{Swin}), they are fused to form a combined feature vector. This fusion constructs a more discriminative representation for grading. Here, the feature fusion is represented as:

$$f_{\text{fusion}} = \text{Concat}(f_{\text{ResNet}}, f_{\text{Swin}}) \quad (10)$$

Such a complementary fusion allows the model to consider both fine-grained biomarkers (e.g., microaneurysms) and retina-wide pathological patterns (e.g., neovascularization) to benefit the separation between closely related classes (Mild vs. Moderate DR).

If W_c and b_c are denoted as the classifier weights and bias, $\hat{y}$ is the predicted probability distribution, then the fused representation is passed through a fully connected classification head defined as:

$$\hat{y} = \text{Softmax}(W_c \text{fusion} + b_c) \quad (11)$$

To minimize the severe class imbalance in the dataset, the Focal Loss function is used during training, which increases the learning emphasis on the minority and misclassified. The loss function is expressed as:

$$L_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (12)$$

where, L_{focal} represents the loss function, α_t denoted as the weighted factor, p_t is the model's predicted probability for the true class t and γ is the focusing strength.

This loss highlights misclassified images, thereby improving the model's sensitivity to minority classes, such as the Mild and Moderate classes in our case.

AdamW optimizer is used in the training process with validation accuracy scheduling. Two-fold stratified cross-validation was used to

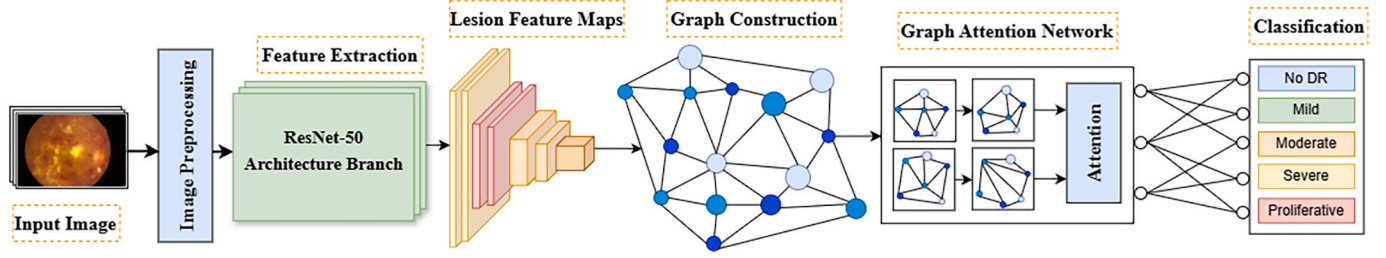


Fig. 5. Simplified ResNet-GAT Architecture Overview.

generalize the results. In the process of inference, a group of fold models is used to predict the end-of-class probabilities, which improves stability and has a lower variance.

Model interpretability is ensured using Grad-CAM, which is applied to the convolutional (ResNet) branch:

$$L_{Grad-CAM} = ReLU\left(\sum_k a_k A^k\right) \quad (13)$$

Fig. 6 illustrates the overall framework of the proposed hybrid structure, where the ResNet-50 is used to extract local retinal fine-grained features, and the Swin Transformer is used to extract global contextual features. All these complementary features combined promote the ability of the model to carry out effective and accurate detection and grading of Diabetic Retinopathy.

4. Experiment

This section outlines the datasets, data preprocessing plan, experimental procedures, evaluation measures, and implementation that will be applied to this study. The experimental study is designed in such a way that it is fair, reproducible, and robust in binary and multiclass diagnostic retinopathy of diabetes (DR) classification.

4.1. Dataset

The main data set to be used in training and evaluation of the model is the APTOS 2019 Blindness Detection dataset (Maggie and Sohler, 2019). It is composed of color retinal fundus pictures obtained in

different imaging conditions and reviewed by clinical scholars. All the pictures are marked as per the 5 levels of DR severity: No DR (0), Mild (1), Moderate (2), Severe (3), and Proliferative DR (4).

There is a significant imbalance in classes, with most of the samples in the No DR and Moderate DR. The imbalance is in accordance with the real-world screening situations and renders the dataset especially adequate to assess the resilience of automated DR classification systems. The dataset is applied in two configurations to help with several clinical use cases:

1. Binary classification: No DR vs. DR (classes 1–4 combined).
2. Multiclass: Five-class grading of the severity of the DR.

In Fig. 7, the APTOS 2019 dataset's binary and multi-class distribution is shown.

Fig. 8 illustrates the percentages of multi-class class distribution of the APTOS 2019 dataset.

To test the generalization of models on different datasets, further experiments on inferences are undertaken with the Indian Diabetic Retinopathy Image Dataset (IDRiD) (Prasanna et al., 2018). IDRiD comprises retinal fundus images of high resolution obtained in a controlled clinical setting and marked by professional ophthalmologists. Unlike APTOS, IDRiD provides controlled conditions of image acquisition, high-quality annotations, and singular lesion-level data.

In the study, IDRiD is applied in testing and inference only. None of the samples of IDRiD are training or validation. Strict separation allows us to evaluate cross-dataset robustness and sensitivity to domain shift due to variation in population demographics, imaging devices, and

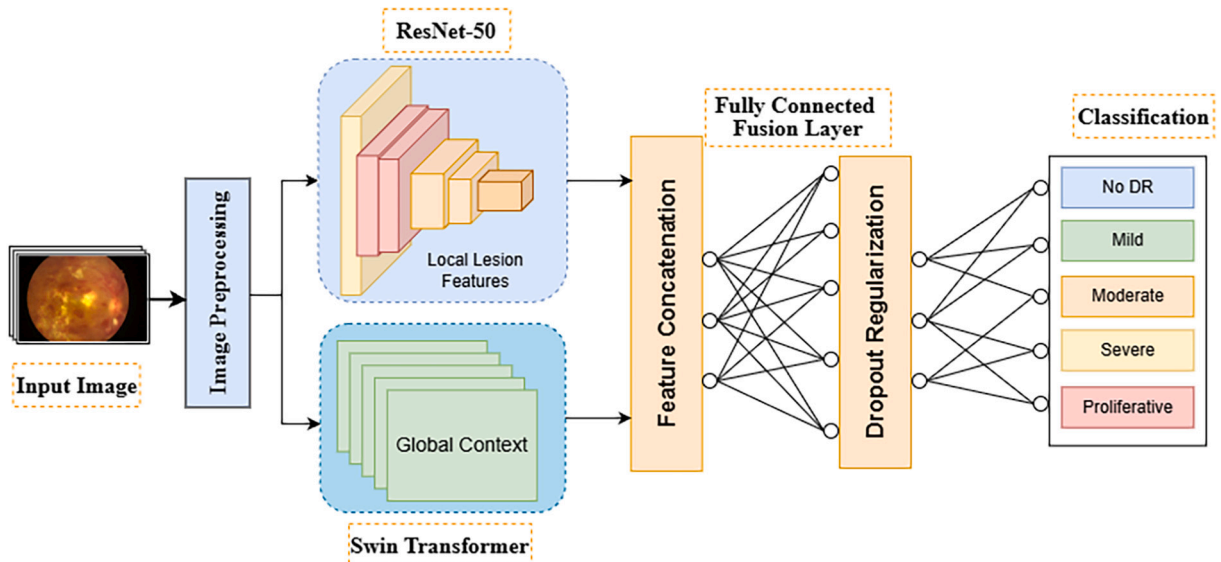


Fig. 6. Architecture of the proposed hybrid model in terms of feature fusion. The local lesion-level characteristics obtained by the ResNet-50 and the global contextual ones by the Swin Transformer are mixed to obtain a whole feature, which is then sent to a fully connected classification head to predict diabetic retinopathy.

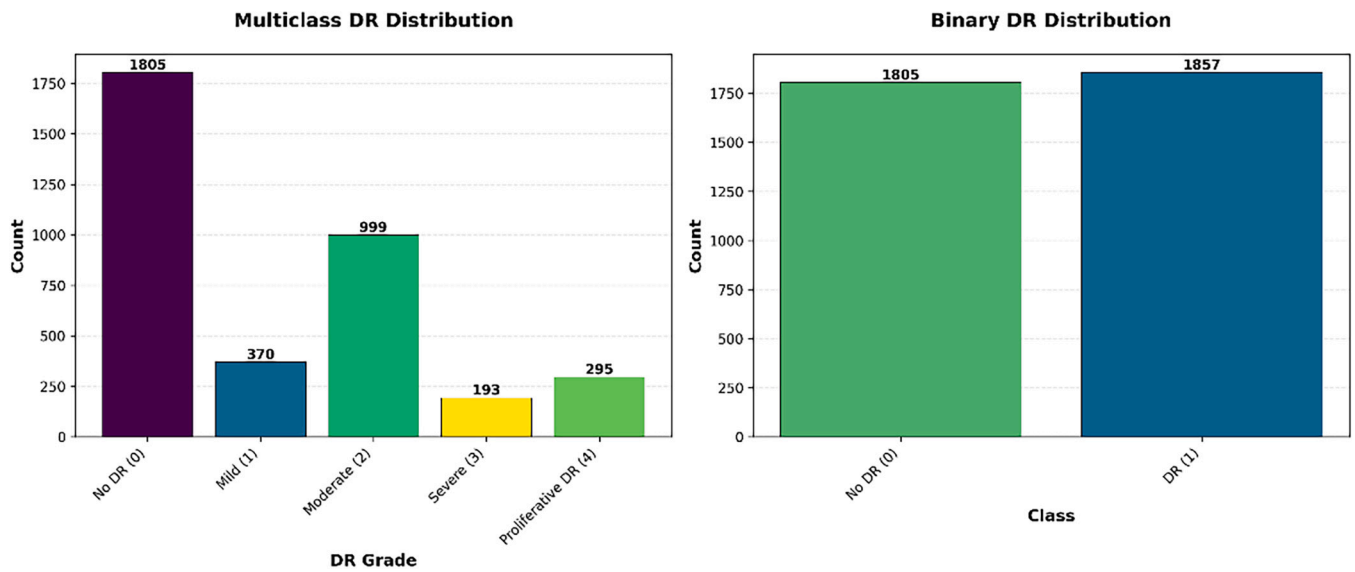


Fig. 7. The class distribution of the APTOS 2019 dataset. On the left side Multi-class five-stage class distribution of the dataset and on the right side, the Binary class distribution is shown, where Class 0 represents No DR and classes 1–4 represent DR.

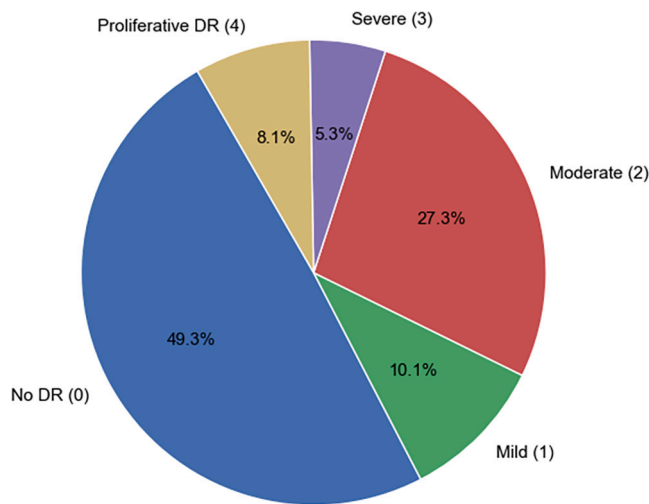


Fig. 8. Percentage of Class Distribution in APTOS 2019 Dataset.

acquisition protocols.

4.2. Data preprocessing

There is a great variation in fundus images since the optics of the camera being used, the light condition, and the position of the patient are different. To achieve this variability minimization and, at the same time, maintain diagnostically relevant features, a standardized pre-processing pipeline is used across all datasets.

Each image is resized to the standardized dimensions of 224×224 pixels, so that it is compatible with ResNet-50 and Swin Transformer models. Tract of Intensity to level out training and decrease response to the changes in illumination. The contrast is enhanced with the help of Contrast-Limited Adaptive Histogram Equalization (CLAHE), which enhances the visibility of vessels and lesions without increasing the intensity of the background noise. Model variants are compared fairly since all of them are processed by the preprocessing pipeline in the same way.

Fig. 9 represents the original and corresponding pre-processed retinal fundus image sample.

4.3. Data augmentation

Data augmentation is only applied to the training set to prevent

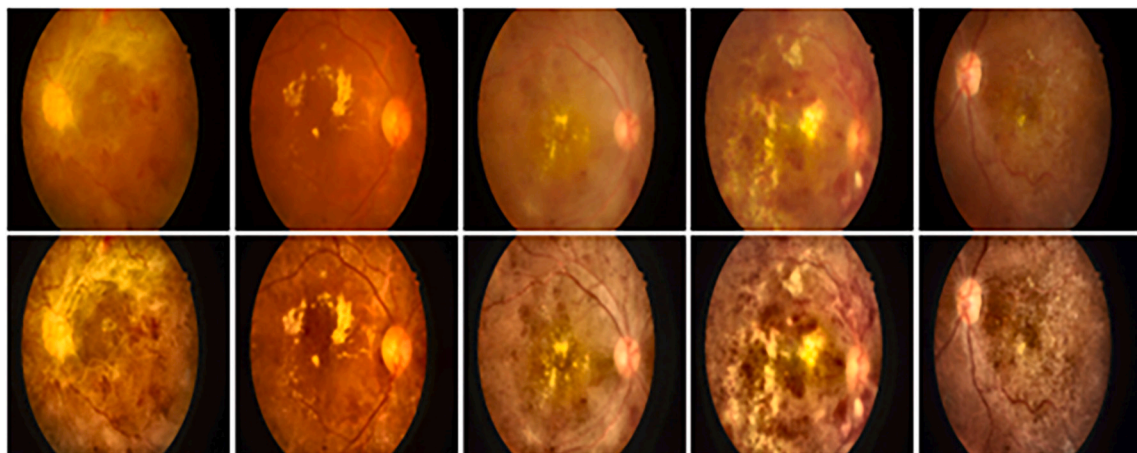


Fig. 9. The preprocessing pipeline showing the impact of normalization and CLAHE on lesion contrast and visibility in fundus images.

overfitting, as well as to alleviate class imbalance. An operation based on augmentation is chosen to maintain clinical realism but augment the data diversity. The augmentations that are applied are: horizontal and vertical flipping, randomly, small-angle rotations, jittering in brightness and contrast, and weak Gaussian noise injection. Validation or test sets are not augmented, avoiding data leaking and making the performance estimation unbiased.

4.4. Experimental protocol

All experiments on the APTOS dataset make use of a stratified two-fold cross-validation approach. Stratification helps maintain the relative distribution of classes within every fold, which is essential in an imbalanced data set like DR severity grading. For each fold, one of these subsets is trained on, and the other subset is to be used in validation and testing.

All architectures (ResNet-50, Swin Transformer, ResNet-GAT, and the hybrid model) reuse the same folds, such that the variation in performance due to the architectural differences is instead due to the data splits.

Binary and multiclass experiments are independently run with the same preprocessing and training parameters.

4.5. Evaluation metrics

The proposed hybrid model was evaluated using multiple statistical metrics commonly used in medical image classification. The evaluation metrics are Accuracy, Precision, Recall, F1-Score, and Quadratic Weighted Kappa (QWK), the official evaluation metric of the APTOS 2019 dataset, which measures the inter-class agreement between the ground truth and model prediction. It is expressed as:

$$k = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}} \quad (14)$$

Here w_{ij} , weight matrix defined as:

$$w_{ij} = \frac{(i-j)^2}{(K-1)^2} \quad (15)$$

where E_{ij} refers to the expected agreement matrix, O_{ij} is the observed agreement. $k = 1$ is the perfect agreement and $k = 0$ represents the random prediction.

For DR grading in this model, this mathematical formula ensures quantitative insight into model reliability and clinical relevance.

4.6. Experimental details

The PyTorch deep learning framework is used to implement all experiments in it. The environment used to conduct model training and evaluation is a Kaggle Notebook, and dual NVIDIA Tesla T4 GPUs (T4 × 2) supplied by the Kaggle platform are used. The setup allows both convolutional and transformer-based architectures to be trained efficiently and consistently across all experimental runs.

The AdamW optimizer is used to perform training, which is very appropriate with deep convolutional and transformer-based models, as it starts by formulating decoupled weight decay. A learning rate scheduling policy is also used to decrease the learning rate progressively over training, which encourages stable convergence. Early validation loss stopping is used to avoid overfitting.

In order to provide reproducibility, it is fixed with random seeds in all experiments. All variants of the model, such as ResNet-50, Swin Transformer, ResNet-GAT, and the proposed hybrid architecture, use the same preprocessing pipeline, optimizer configuration, learning rate schedule, and training protocol.

Focal loss is used to tackle the class imbalance in the diabetic retinopathy datasets, which will prioritize the hard-to-classify domain, and

the prevalent classes will be diminished. All experiments have the same hyperparameters (learning rate, batch size, focal loss parameters, and training epochs) to make a fair and unbiased comparison.

4.7. Experiment output

To test the model's performance, all the baseline models were stacked and created a stacked-ensemble model, and it was directly applied to the IDRiD dataset. This experiment shows the robustness of the stacked-ensemble model on data with different acquisition settings, illumination, and population.

Fig. 10 demonstrates that the ensemble model maintains consistent severity grading across all five DR classes and achieves a reasonable confidence score on unseen data under different illumination conditions. Though the Mild and Moderate classes show comparatively lower confidence scores, this is clinically acceptable due to overlapping lesion characteristics.

In Fig. 11, the prediction distribution shows an even understanding of the Mild and Moderate DR classes, but shows little understanding of Severe and Proliferative, reflecting the characteristics of the dataset. The confidence boxplots show that the median confidence for No DR and Proliferative DR is higher, while the middle classes have more variance, which can be interpreted as indicating more uncertainty for the other classes.

In Fig. 12, the model shows the best discrimination for No DR (0.9420) and Proliferative DR (0.8737), which shows strong separation at the extreme severity level. The Moderate discrimination performance for Mild and Moderate DR is due to the proximity and inherent difficulty in differentiating adjacent stages, particularly under domain shift; the micro-average ROC-AUC of 0.7921 shows stable multi-class separability without any re-training.

4.8. Zero shot external validation

Cross-dataset generalization was achieved by directly applying the proposed Hybrid ResNet-Swin model to the IDRiD dataset (455 images) without any fine-tuning or retraining; none of the IDRiD samples were used for Training, validation, or threshold selection. Because the acquisition device differs between IDRiD and APTOS, it is a hard test of the domain's robustness when the distributions of resolution, population, and grade are known. Quantitative: The summary of results is presented in Table 3.

In this zero-shot scenario, the model maintained good ordinal consistency, achieving 97% accuracy on quadratic weighted kappa. Kappa value was 0.694, and the correctness rate was 88.6% within one severity level, meaning that it did not make many mistakes. Most of them were within adjacent grades as opposed to clinically distant grades (Fig. 13, confusion matrix). As would be predicted with a large domain shift, exact 5-class accuracy dropped to 43.3% (mainly due to the drop in accuracy for class 2).

The differences between the two are statistically significant not only in the minority Mild and Severe classes, but also in the significantly different grade distribution of IDRiD. Importantly, the AUC of the clinically critical referable-DR decision (grade ≥ 2) was preserved at high levels. The specificity, almost 0.959, and the precision, 97.4% (cases identified as "referable" were almost all), were very high. Disease discrimination was likewise strong (AUC 0.942), as was No-DR. Together, these results, without any target-domain adaptation, show that the model retains screening-level utility on a few datasets, as shown by results in a fully independent dataset.

The accuracy on the exact grade is not as good as the other classes, and the sensitivity is low for the classes Mild and Severe under domain shift, revealing that fine-grained cross-domain grading is not so easy; these classes are also the scarcest. Subject to inter-observer variation. This encourages light domain adaptation, and further external work is referred to as cohorts and not as domain-invariant grading (Fig. 14).

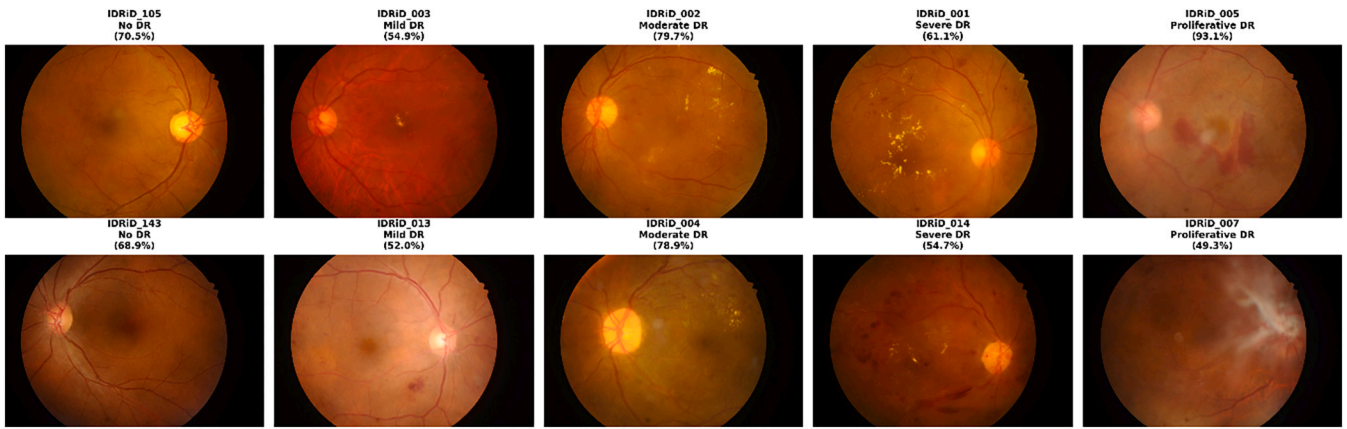


Fig. 10. IDRiD fundus images with predicted DR classes (Class 0 to Class 4) with confidence scores using the Stacked-Ensemble Model.

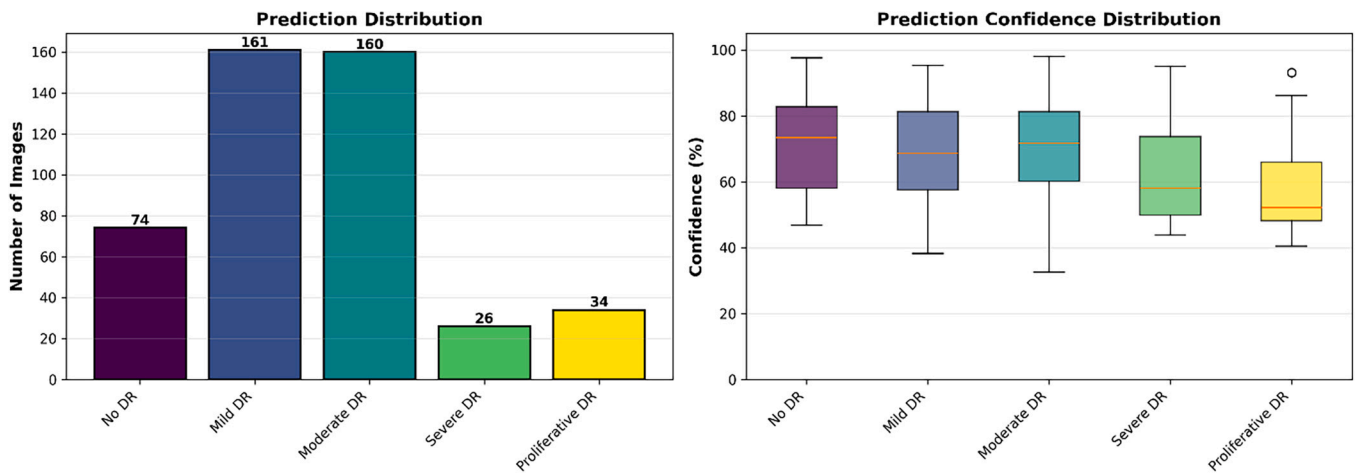


Fig. 11. Distribution of predicted DR classes on the IDRiD dataset and confidence distribution across all five classes.

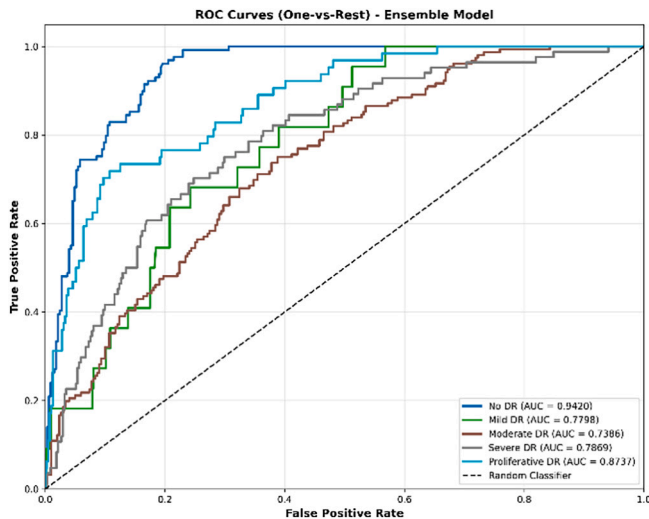


Fig. 12. One vs Rest ROC Curve on the IDRiD Dataset using the Stacked-Ensemble Model.

Thus, the overall summary is that the model achieved significant ordinal accuracy on the IDRiD dataset, applied to this dataset without any fine-tuning. The agreement (QWK: 0.69, 88.6% within one grade) and high discrimination (AUC: 0.96) with high referable-DR were

Table 3

Per-Class performance on IDRiD Dataset (Zero-shot).

Class	Support	Precision	Recall	F1-score	AUC
No DR	129	0.838	0.481	0.611	0.942
Mild	22	0.093	0.682	0.164	0.780
Moderate	156	0.513	0.526	0.519	0.739
Severe	84	0.577	0.179	0.273	0.787
Proliferative DR	64	0.677	0.359	0.469	0.874

achieved. Supports its cross-dataset robustness in screening. Further, the proposed model's performance on the primary dataset is discussed in the next section.

5. Result and discussion

This section provides the quantitative as well as the qualitative analysis of the proposed framework on the primary dataset (APTOS 2019). The results of the individual architecture-based approaches included ResNet-50, Swin Transformer, and ResNet-GAT-s that were initially evaluated in binary and multi-class diabetic retinopathy (DR) classification tasks. After that, the novel hybrid ResNet-50 + Swin Transformer model is tested and compared to the baseline architectures. Lastly, there are ablation studies, which are a systematic study of the role of every architectural element and design option.

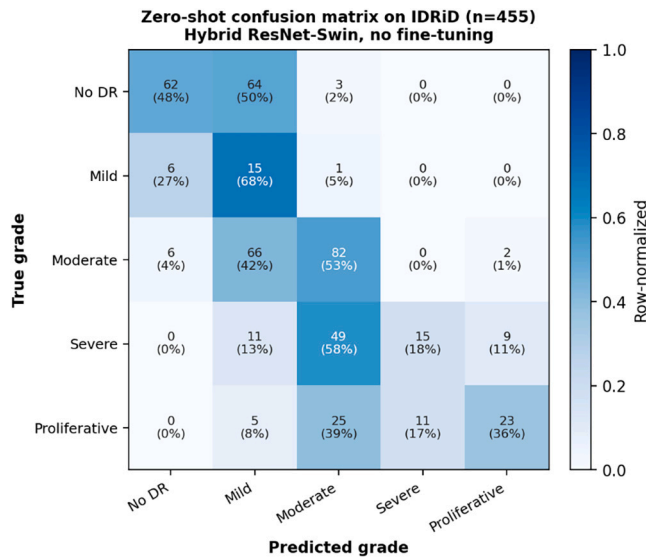


Fig. 13. Zero-shot Confusion Matrix on IDRiD Dataset using Hybrid ResNet-Swin Model.

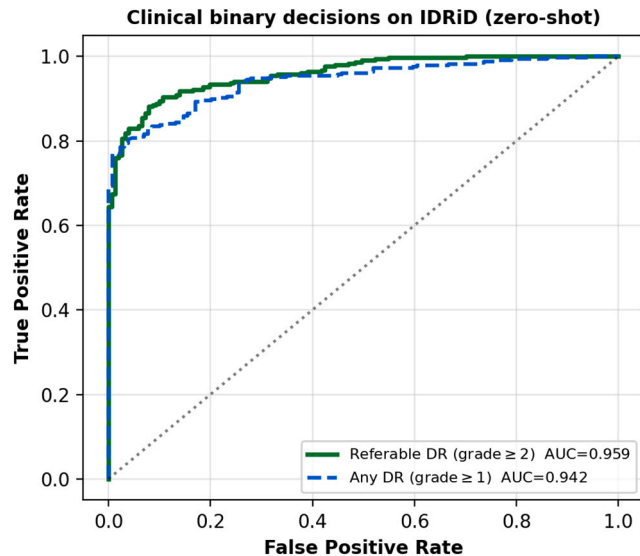


Fig. 14. Zero-shot Clinical Binary Decisions on IDRiD dataset using Hybrid ResNet-Swin Model.

5.1. Binary DR classification

Experiments of binary classification are run to differentiate No DR and DR cases. The given task is associated with practical screening situations when the key task is to determine patients who need additional clinical evaluation. Binary classification is experimented with using the proposed Hybrid Swin-ResNet Model.

The class distribution is given in Table 4.

The proposed two-fold ensemble hybrid model achieved an accuracy of 98.09%, precision of 97.65%, recall of 98.60%, F1 score of 98.12%, and a Quadratic Weighted Kappa (QWK) of 0.9618, indicating strong

Table 4

Class distribution table for Binary Classification.

Class	Image count
No DR (Class 0)	1805
DR (Class 1)	1857

agreement with expert ophthalmologist annotations. The Binary classification metrics are given in Table 5.

The binary classification fold-wise training and validation accuracy curve and QWK progress curve are shown in Fig. 15.

The best-fold-wise validation accuracy and QWK, and the confusion matrix are shown in Fig. 16.

5.2. Multiclass DR classification

5.2.1. Severity grading of DR with hybrid Swin-ResNet model

In the multi-class grading experiment, the proposed method classified the severity into five classes. They are: No DR (0), Mild (1), Moderate (2), Severe (3), and Proliferative DR (4). The dataset was imbalanced, containing 49.3% of the images of class 0 or No DR. Stratified sampling helped ensure each fold had equal class representation.

The proposed ResNet50 and Swin Transformer-based hybrid model achieved an overall accuracy of 94.89%, an average F1-score of 94.86%, and average precision of 94.94%. The Quadratic Weighted Kappa score of 0.9726 clearly ensures the reliability of the hybrid model for predicting multi-class Diabetic Retinopathy.

The class-wise performance metrics for multi-class DR classification are shown in Table 6.

The loss function curve per epoch for fold 1 and fold 2 is shown in Fig. 17.

The development of training in the two folds indicates good and fast reduction in the Training Loss with very low values achieved by the end of the 20th epoch in both folds. Nonetheless, the Validation Loss indicates that there is a possibility of overfitting in Fold 1, due to the fact that the difference between the training and validation loss increases dramatically past the 10th epoch. Conversely, Fold 2 has a slightly smaller Validation Loss, more fluctuation, and more plateaus (at approximately 0.18–0.20), showing that Fold 2 has different generalization behavior between data splits. The multi-class grading evaluation metrics accuracy curve and are shown in Fig. 18.

The multi-class assessment through the two folds shows a strong learning trend in the model. The Training Loss reduces well in both folds, meaning that the model is learning the training data fast.

Nonetheless, when the Validation Loss and Validation Accuracy are analyzed, it is indicative of a bit of overfitting. In particular, the difference between the training and validation loss scales, and the Validation Accuracy curves become saturated at a much lower value than the almost optimal training performance.

Most importantly, the Quadratic Weighted Kappa (QWK) curves show that even with this slight overfitting, the model is very efficient in the main task of severity grading, as the QWK values, which gauge agreement with human grading, are achieved at very high levels and are not dropping as the training goes on.

Class-wise evaluation metrics plot and the confusion matrix are shown in Fig. 19.

The results of the multi-class evaluation, which are provided in the three figures, indicate that the model is overall effective in grading Diabetic Retinopathy. The loss function curves (Fig. 16) indicate a stable Validation Loss (Fold 1: ≈ 0.17 , Fold 2: ≈ 0.18) despite a large decrease in the Training Loss, which points to some overfitting. But the most

Table 5

Average accuracy and evaluation metrics for Binary classification on the APTOS dataset.

Metric	Fold 1	Fold 2	Average
Accuracy	0.9815	0.9803	0.9809
Precision	0.9781	0.9750	0.9765
Recall	0.9864	0.9856	0.9860
F1-score	0.9817	0.9808	0.9812
QWK	0.9623	0.9613	0.9618

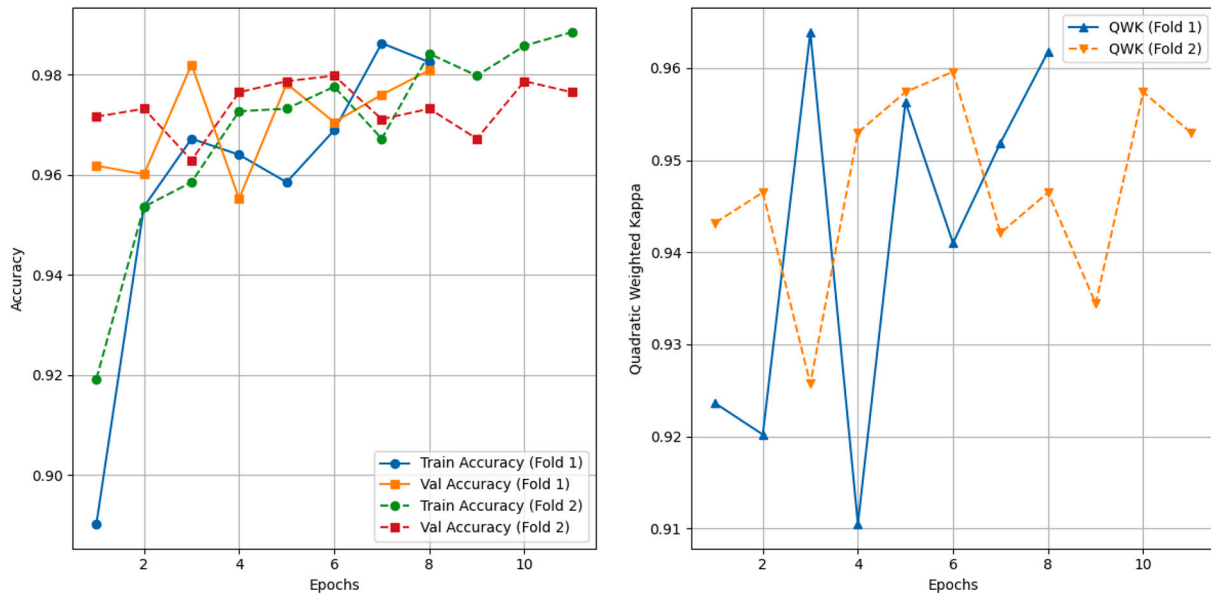


Fig. 15. This figure represents fold-wise accuracy curves of the training and validation set and the Quadratic Weighted Kappa.

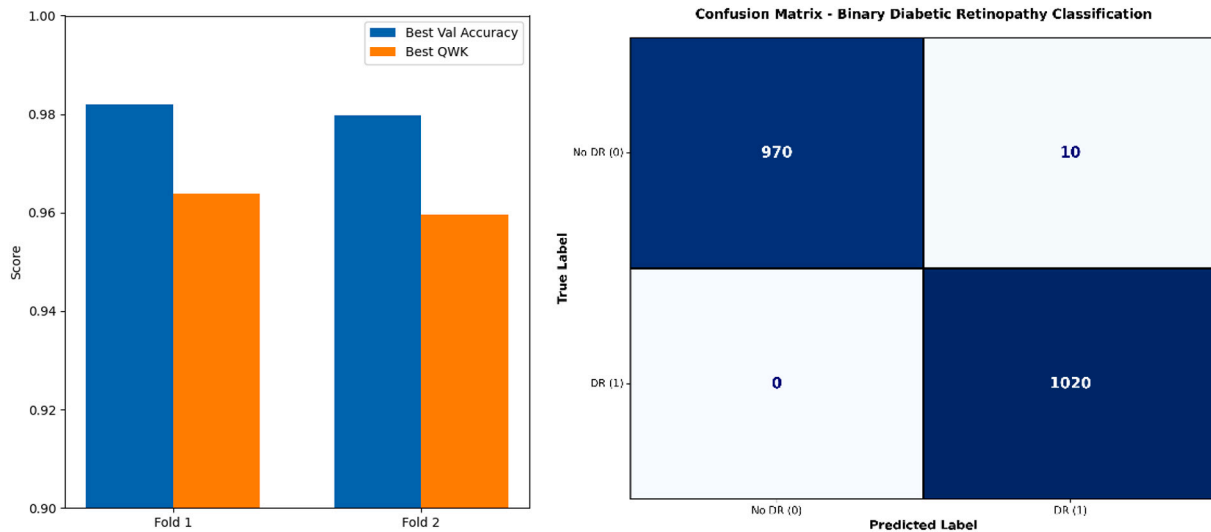


Fig. 16. The best-fold-wise validation accuracy and QWK, and the confusion matrix for the binary classification.

Table 6

Class-wise performance metrics for multi-class DR classification.

Class	Precision	Recall	F1-score
No DR	0.9950	0.9867	0.9908
Mild	0.8852	0.8541	0.8693
Moderate	0.9047	0.9690	0.9357
Severe	0.9118	0.8031	0.8540
Proliferative DR	0.9273	0.8644	0.8947
Weighted avg	0.9494	0.9489	0.9486

important measure of evaluation, Quadratic Weighted Kappa (QWK) (Fig. 17), attains high and steady values (Fold 1: ≈ 0.90 , Fold 2: ≈ 0.89), and this suggests that severity is in good agreement with the human grading. The classification-based metrics (Fig. 18) demonstrate that the category of No DR (Class 0) has outstanding performance with Precision, Recall, and F1-score close to 1.00. The highly critical stage of the performance is also good, with a precision of 0.92 and an F1-score of 0.90 for the Proliferative (Class 4) stage. The Confusion Matrix validates

correct diagnosis, especially those of the No DR and moderate stages.

5.2.2. Severity grading of DR with ResNet-GAT model

To examine the role played by the relational lesion modeling, a multi-class diabetic retinopathy severity grading experiment was performed with the proposed ResNet-Graph Attention Network (ResNet-GAT) architecture. Just like the hybrid model, the task was to categorize fundus images as one of five severity levels of No DR (0), Mild (1), Moderate (2), Severe (3), and Proliferative DR (4). The identical imbalanced dataset distribution was also used, and about 49.3% of the samples fell in the No DR category. Stratified sampling was used to maintain a consistent sampling of the classes in each fold.

ResNet-GAT model takes advantage of convolutional feature representation to obtain lesion-level representations, and the graph-based attention to explicitly represent inter-lesion connections and local dependencies. The design is especially driven by the clinical experience that intermediate stages of the DR are usually based on the co-existence and spatial distribution of several lesions as opposed to isolated pathological findings.

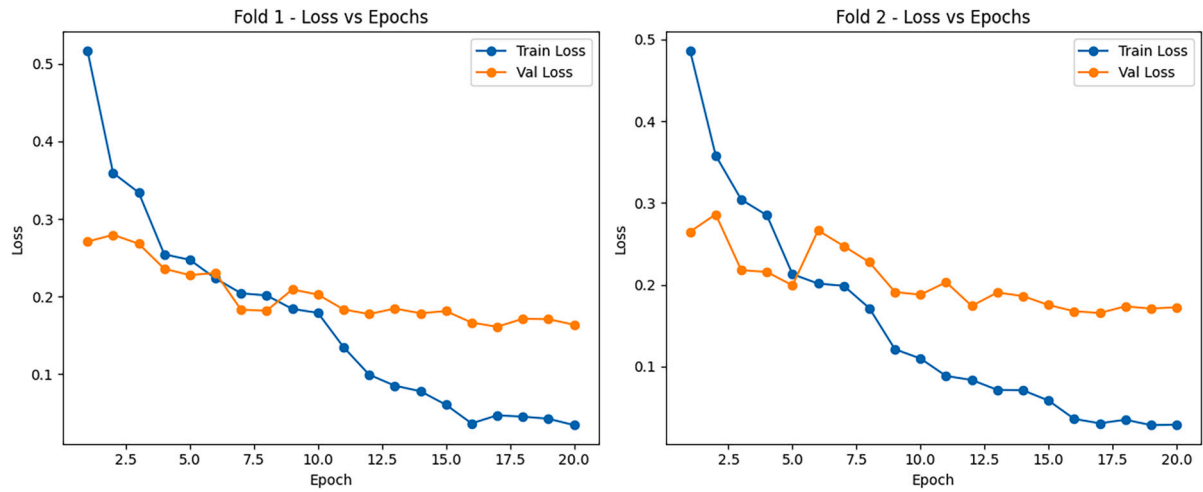


Fig. 17. Loss function curves for fold 1 and fold 2 for Multi-Class.

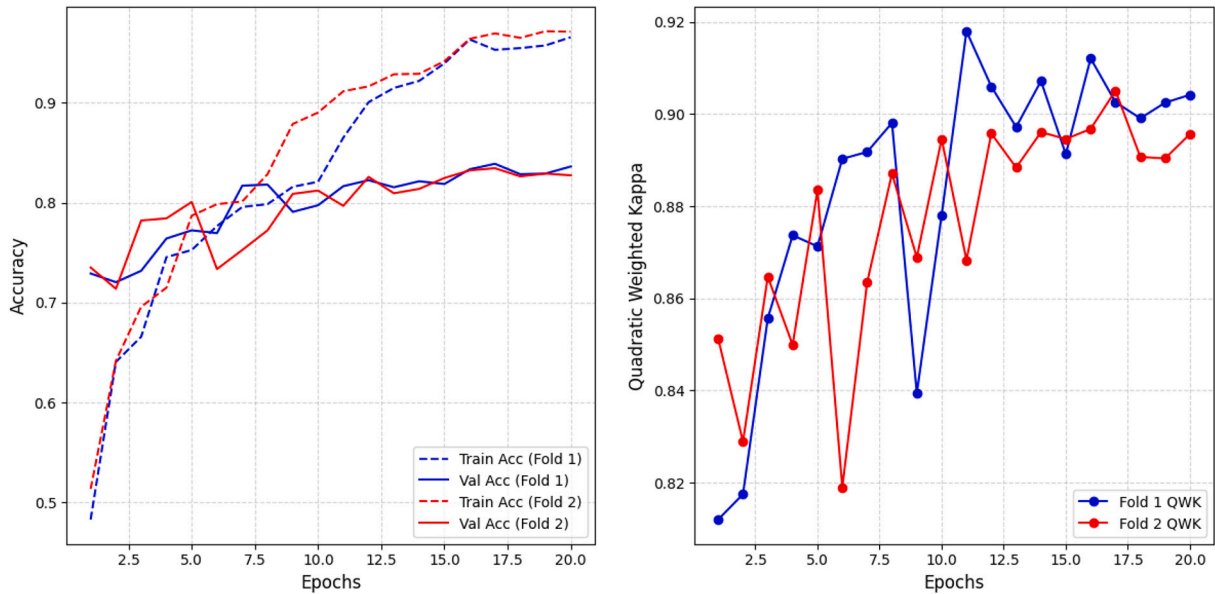


Fig. 18. Multi-class accuracy and QWK progress curve.

One of the models obtained the overall competitive performance, having steady accuracy and good agreement with the ground-truth classes, especially in classes such as Mild and Moderate DR. Although the general accuracy and QWK scores are slightly lower than the hybrid Swin-ResNet model, the ResNet-GAT model showed better consistency in terms of the classes than the plain ResNet-50 model, which confirmed the usefulness of relational reasoning. The overall test accuracy is 80%, and QWK is 85.68%.

The class-wise performance metrics for multi-class DR classification using the ResNet-GAT model are shown in Table 7.

In Fig. 20, the metrics of class-wise evaluation and the confusion matrix of ResNet-GAT have been included. The confusion matrix shows that the model is successful, especially in the differentiation of No DR and Moderate DR, and few misclassifications between the two stages are made as opposed to the CNN-only baseline. Misclassification errors occur mostly between the next severity levels (Mild-Moderate and Severe-Proliferative), which are clinically understandable, since there is a similarity in the appearance of the lesions.

Fig. 21 shows the ROC curves on the test set of the APTOS 2019 dataset.

Fig. 22 shows the precision-recall curves on the test set on the APTOS 2019 dataset.

Overall, the ResNet-GAT results show that explicit modeling of relational lesions is better at severity discrimination, particularly in clinically ambiguous intermediate phases. Nevertheless, its lack of explicit global retinal context restricts its capacity to capture large-scale pathological patterns, which is why it has poor performance in comparison with the hybrid Swin-ResNet model. These results confirm that relational reasoning is helpful but most effective when combined with global contextual modeling.

5.3. Ablation study and comparative model analysis

A systematic ablation study was performed to quantify each element of proposed framework and justify the architectural and design decisions. We report two complementary analyses: (i) an architectural ablation where we test each backbone individually against the fused model by removing one training ingredient at a time; and (ii) a component ablation on the proposed hybrid model to remove one training ingredient at a time, keeping the rest fixed, with the focal loss

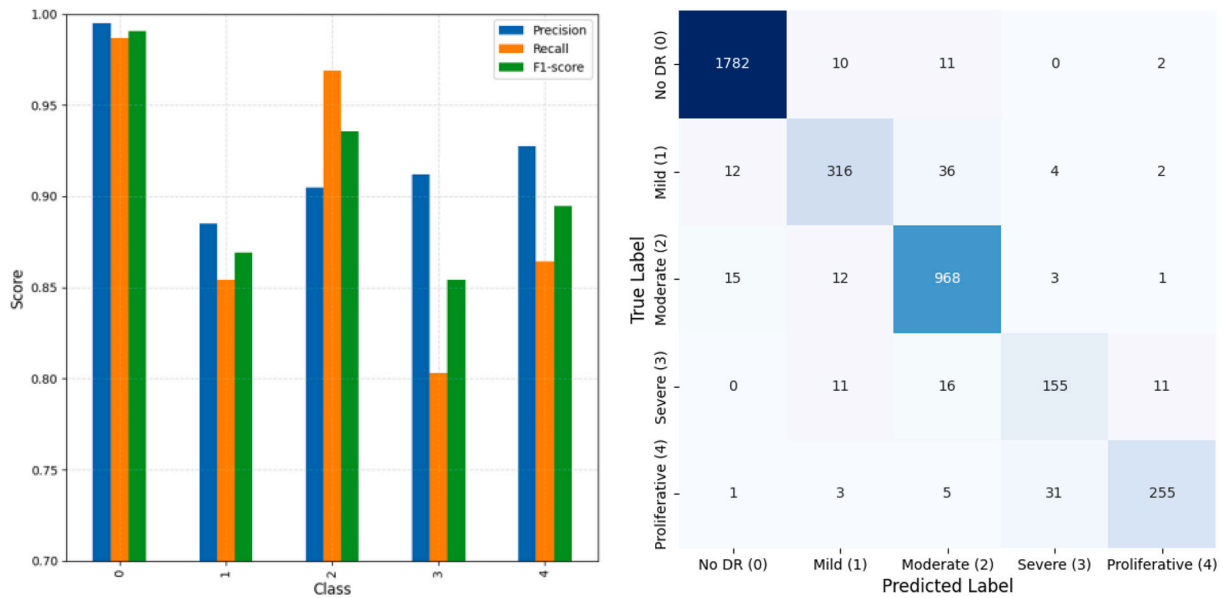


Fig. 19. Precision, Recall, F1-Score, and Confusion Matrix for multi-class grading.

Table 7

Class-wise performance metrics for multi-class DR classification using ResNet-GAT.

Class	Precision	Recall	F1-score
No DR	0.96	0.98	0.97
Mild	0.56	0.70	0.62
Moderate	0.75	0.69	0.72
Severe	0.31	0.38	0.34
Proliferative DR	0.61	0.43	0.51
Weighted avg	0.80	0.80	0.80

and the CLAHE preprocessing. All experiments were performed with the same dataset splitting, preprocessing pipeline (except ablated), loss function (except ablated), optimizer and two-fold training schedule as described in Section 4, therefore the differences in performance scores correspond only to the ablated factor. The two-fold ensemble is reported values, in line with the headline multiclass results. The results obtained are summarized in Table 8 below, which focuses on: (1) the contribution of the local and global feature modeling, and their fusion; (2) the effect of the focal loss and CLAHE preprocessing on minority-class severity grading; and (3) the trade-off between performance and model complexity.

The architecture was left unchanged, and the only change was to replace the focal loss with a standard cross-entropy loss, which lowered accuracy from 0.9489 to 0.9001, and the macro F1-score from 0.9089 to 0.8241, while the quadratic weighted kappa dropped from 0.9726 to 0.9435. The degradation was more pronounced in the minority and intermediate severity classes (Mild DR and Severe DR) with a decrease in per-class F1-scores to 0.76 and 0.71 respectively. This shows that the focal loss indeed achieves the desired effect of improving the grading of under-represented classes by down weighting the easy samples in the majority classes, and putting more effort into training classes that contain hard, clinically relevant samples. Table 8 represents the ablation study of the proposed framework.

The omission of CLAHE contrast enhancement from the preprocessing pipeline, as well as cropping, sharpening, and retaining all of the training parameters, resulted in a drop in accuracy from 0.9489 to 0.9181; the macro F1-score decreased from 0.9089 to 0.8544, and the quadratic weighted kappa dropped from 0.9726 to 0.9576. The steady decrease in all parameters provides further evidence that local contrast equalization enhances the visibility of subtle retinal lesions

(microaneurysms, hemorrhages, and hard exudates) and thus aids more accurate severity classification.

The ResNet-50 baseline has shown a good basis for diabetic retinopathy classification since it can produce fine-grained lesion-level features. Nevertheless, when it comes to multiclass grading of severity, especially at intermediate levels, it becomes saturated because of the lack of global retinal context.

The Swin Transformer is better in its accuracy and F1-score due to the model of long-range dependencies and hierarchical retinal structure. This validates the fact that global contextual information is very important in severity grading. Transformer-only models are, however, more complex to calculate and less sensitive to fine lesion-scale patterns in early-stage DR.

The ResNet-GAT model introduces relational reasoning between lesion regions, which leads to greater stability in classification, particularly for the Mild and Moderate DR classes. Although this is an improvement, its performance is not as good as that of the hybrid architecture because of the absence of explicit global contextual modeling.

The proposed hybrid ResNet-50 + Swin Transformer model records the highest results in all indicators due to the integration of local sensitivity to lesions and global contextual awareness. Even though this model contains more parameters, the performance improvement is uniform and well worth the compensatory quality of the fused representations. Notably, the increase in accuracy and F1-score is attained without having more complex attention fusion modules, which keeps the architectures simple.

5.4. Comparative model analysis with recent state-of-the-art methods

A comparison of the proposed framework with the recent DR classification methods assessed on the APTOS 2019 data set is shown in Table 9. However, due to the different evaluation protocols used in previous studies (hold-out validation, public/private Kaggle test splits, and k-fold cross-validation), along with the various classification objectives (binary screening to five severity classes), it is important to consider methodological differences when interpreting the results of the performance.

It is difficult to compare performance directly, due to the different evaluation goals discussed in the reviewed studies. There are several techniques that have been reported to have higher accuracy but lower QWK scores and other techniques that have higher QWK scores with

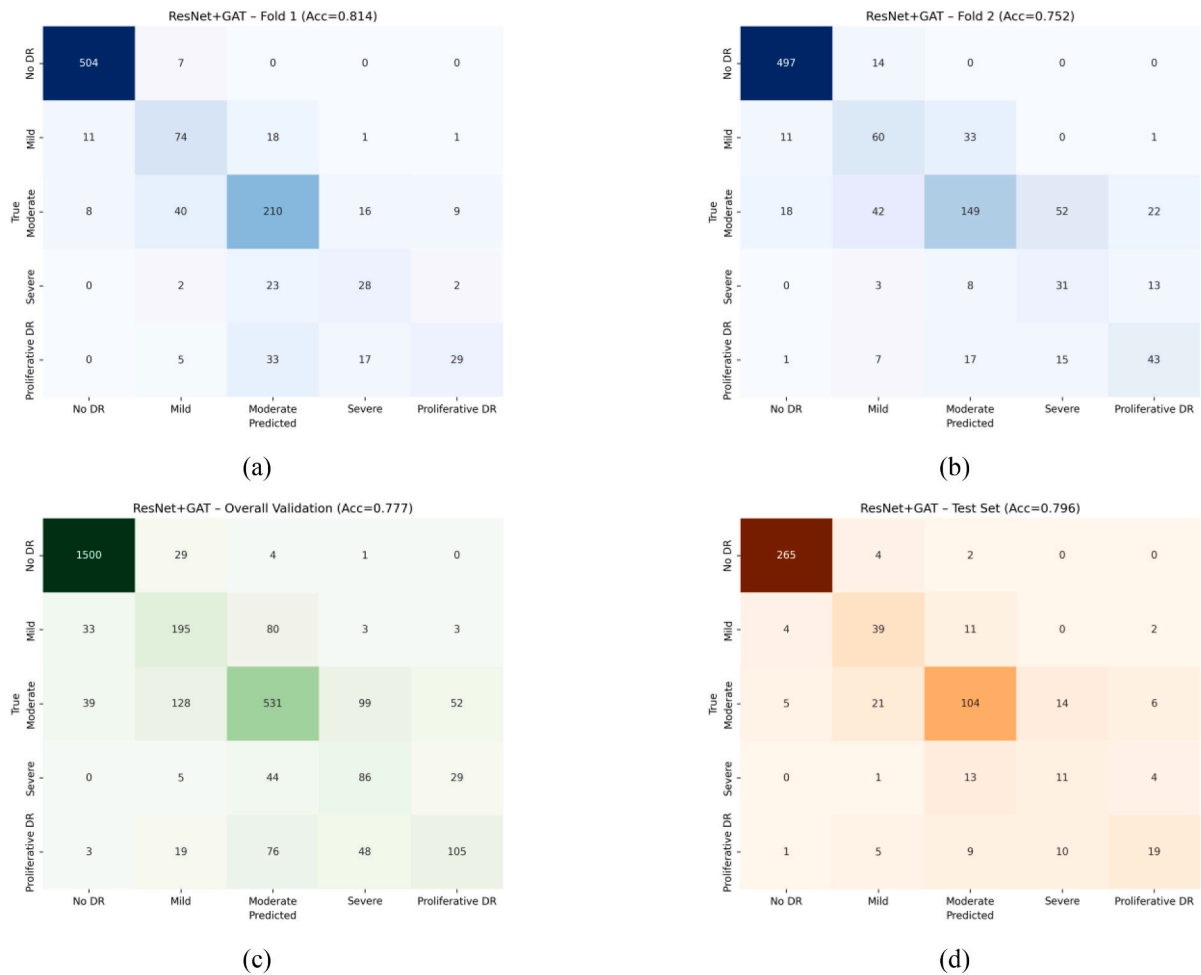


Fig. 20. Confusion Matrix. (a) Fold 1 Confusion Matrix. (b) Fold 2 Confusion Matrix. (c) Overall validation Confusion Matrix. (d) Test Set Confusion Matrix.

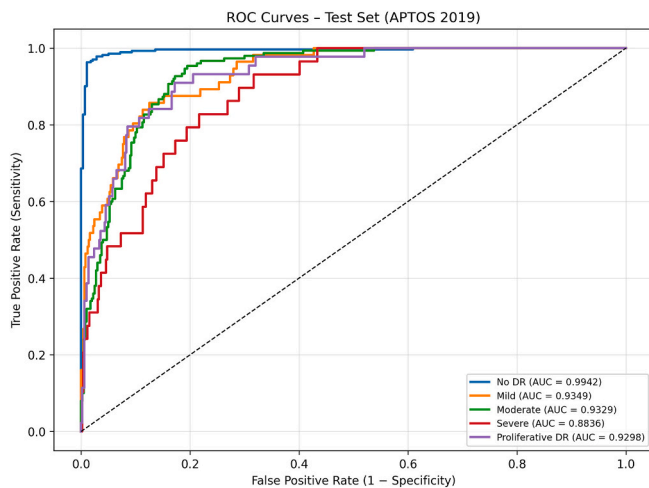


Fig. 21. ROC Curves of ResNet+GAT on the Test Set of APTOS 2019 Dataset.

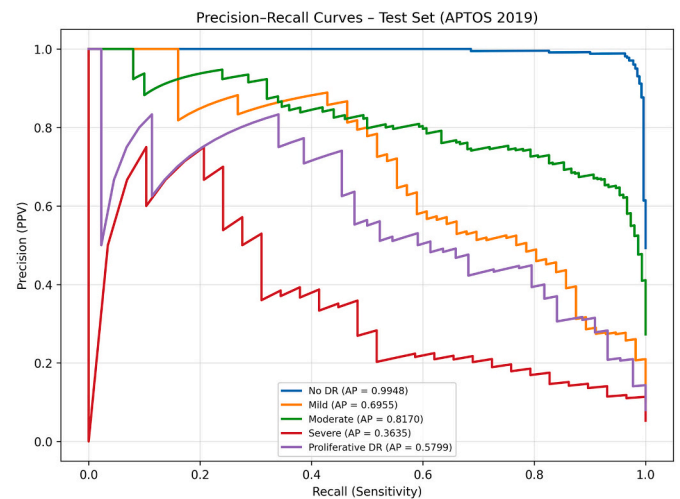


Fig. 22. Precision-Recall Curves of ResNet+GAT on the Test Set.

comparatively low accuracy. Thus, neither of these can be used as the sole criterion of overall superiority, and results must be evaluated based on the task to be performed and the evaluation protocol.

The proposed hybrid framework performs 5-class diabetic retinopathy grading with a Quadratic Weighted Kappa (QWK) score of 0.9726 as presented in Table 9 using stratified cross-validation. This is competitive with the recently published state-of-the-art approaches of

APTOS 2019 and highlights the effectiveness of the proposed architecture in grading with accurate and reliable results for all the grading categories.

5.5. Computational complexity and deployment feasibility

The computational profile of all four architectures is reported in

Table 8
Ablation Study of the Proposed framework.

Model	Accuracy	Macro Avg	Weighted Avg
ResNet-50	0.6726	0.5059	0.6890
Swin Transformer	0.6958	0.5631	0.7049
ResNet-GAT	0.80	0.63	0.80
Proposed hybrid w/o Focal loss (cross-entropy)	0.9001	0.8241	0.8992
Proposed hybrid w/o CLAHE preprocessing	0.9181	0.8544	0.9169
Proposed Hybrid Model (ResNet-Swin)	0.9489	0.9089	0.9486

Table 9
Comparison study and evaluation with recent methods on APTOS 2019 dataset.

Method (author, year)	Approach	Task	Accuracy	QWK/AUC
Mishra & Hanchate, 2020 (Mishra et al., 2020)	Deep CNN	5-class	96.11%	QWK = 0.8981
Bodapati et al., 2021 (Bodapati et al., 2021)	Deep CNN, Feature Aggregation	5-class	84.31%	AUC = 97
Wahab Sait, 2023 (Wahab Sait, 2023)	YOLOv7 + MobileNetV3	5-class	98.0%	QWK = 0.911
Ohri et al., 2024 (Ohri et al., 2024)	Self-supervised ViT	5-class	Not Reported	QWK = 0.8748
S. V. Chilukoti, 2024 (Chilukoti et al., 2024)	Transfer Learning Ensemble	5-class	Not Reported	QWK = 0.967
D. Dharrao, 2025 (Dharrao et al., 2025)	EfficientNetB0 + MSAG	5-class	88.4%	QWK = 0.923
M. Youldash, 2025 (Youldash et al., 2024)	RegNetX080	Binary	98.6%	Not Reported
M. Youldash, 2025 (Youldash et al., 2024)	EfficientNetB3	5-class	85.1%	Not Reported
Proposed Hybrid Model	Hybrid CNN-Transformer	Binary	98.09%	QWK = 0.9618
Proposed Hybrid Model	Hybrid CNN-Transformer	5-class	94.89%	QWK = 0.9726

Table 10
Computational complexity of all four architectures.

Model	Params (M)	FLOPs (G)	GPU (ms/img)	CPU (ms/img)	Peak mem (MB)
ResNet-50	23.52	8.26	6.25	88.79	133.2
Swin-Tiny	27.52	8.74	12.22	124.82	139.6
ResNet-GAT	25.87	8.26	8.61	95.07	142.6
Hybrid ResNet-Swin	110.27	38.60	34.05	452.25	488.5

Table 10, measured using the same protocol: 224×224 input, batch size 1, and a single NVIDIA Tesla T4 GPU; latency is reported as the median over 100 forward passes after 20 warm-up passes. The single-paradigm baselines are also lightweight with peak GPU memory usage of less than 145 MB: ResNet-50 (23.5 M parameters, 6.25 ms/image), Swin-Tiny (27.5 M, 12.22 ms), and ResNet-GAT (25.9 M, 8.61 ms). The reported FLOPs are for the ResNet-50 feature-extraction backbone of ResNet-GAT, since its two graph-attention layers operate on a small 7×7 node graph and impose only a small additional overhead that is not captured by standard FLOP profilers. The most demanding of all four is the proposed hybrid ResNet-Swin model with 110.3 M parameters, 38.6 GFLOPs, and 34.05 ms on GPU (452 ms on CPU) with a peak memory of 489 MB. This equates to approximately a 4–5 $\times$ increase in parameters and a 3–5 $\times$ increase in latency compared to a single backbone, which

still results in a 29 images per second performance on a single GPU with a latency of 34 ms/image, well within the screening requirements of real-time.

There are two operating regimes for deployment. The reported results employ a fold-ensemble which averages predictions across the folds to decrease variance - this increases the inference cost by the number of folds; it is appropriate for offline or server-side grading, where the inference latency of 34 ms/image for a single model is easily achieved by commodity GPUs, as is the model memory footprint of 489 MB. A single-fold hybrid model can be used for point-of-care and low-resource screening at 34 ms/image on GPU and 452 ms/image on CPU, providing real-time throughput without a dedicated accelerator. Where hardware is more limited, the lightweight ResNet-50 or ResNet-GAT branches (around 6–9 ms/image) are available. Hybrids with <145 MB have a good accuracy-cost ratio, and INT8 quantization and structured pruning can additionally shave model size and latency with little loss of accuracy after training. These options combine to create a feasible system for deployable diabetic-retinopathy screening in under-resourced areas.

5.6. Visualization and interpretability

Diabetic retinopathy (DR) screening systems may be deployed in the clinic, not only with high classification accuracy but also with easy interpretability. In the medical context, it is critical to know what parts of the retina determine model choices to be trustworthy and transparent. To overcome this, the Gradient-weighted Class Activation Mapping (Grad-CAM) is used to visualize discriminative regions exploited by the trained models in the inference process.

Fig. 23 displays Grad-CAM images of the hybrid ResNet-50 + Swin Transformer model with the original fundus image, heatmap, and heatmap on the fundus image. The activation maps have always shown an emphasis on the clinically significant pathological areas like hemorrhages and exudates, and have inhibited the background responses. The high correspondence between highlighted areas and visible lesions suggests that model predictions are motivated by significant retinal structures. This value confirms the usefulness of the hybrid architecture to combine locally sensitive CNN results with the overall contextual awareness transformer. The directed but consistent patterns of activation indicate that the feature fusion strategy maintains information at the lesion-level and is more interpretable.

Fig. 24 shows the maps of Grad-CAM abnormality on fundus images with various severities of diabetic retinopathy. Such maps highlight the regional distribution and magnitude of anomalous areas that lead to the forecasts of the model.

In the case of early-stage DR, the areas of activation are limited and localized, which indicates the slight presence of lesions. With progression to more severe levels of disease, Grad-CAM responses become increasingly spatially widespread and excessive, along with the clinical progression of diabetic retinopathy, whereby pathological alterations become diffuse over larger areas of the retina. The figure is added to explain the high Quadratic Weighted Kappa (QWK) scores of the model. As QWK penalizes distant severity misclassifications more intensely, the observed monotonic expansion of attention across severity levels furnishes visual data that the model in question learns an ordinal representation of disease progression, but not classes.

The results of graph-guided abnormality maps that were produced by the ResNet-Graph Attention Network (ResNet-GAT) are presented in Fig. 25. The figure comprises the initial fundus image, the graphical representation of the abnormality in the form of a heatmap, and the overlay with colors assigned to the various lesion-like areas.

The colors red, yellow, and blue represent areas of hemorrhage-like response, exudate-like response, and vessel-dense areas, respectively, in this visualization. The graph-guided map, unlike Grad-CAM, which focuses on discriminative regions according to the importance of features, highlights relational dependencies between lesion regions.

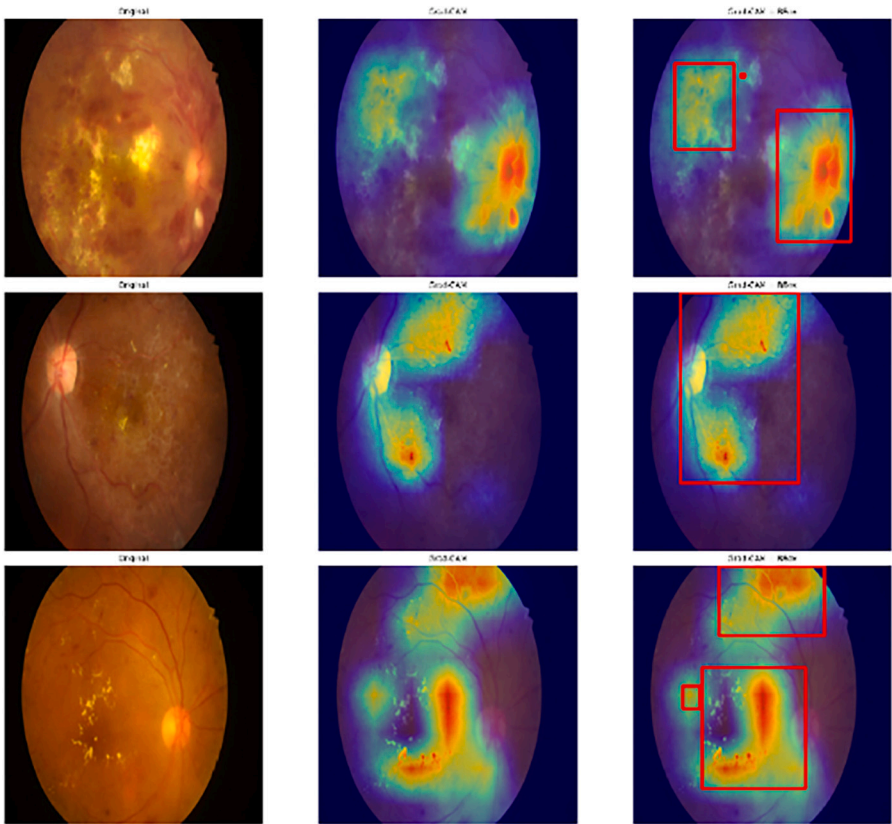


Fig. 23. Grad-CAM visualization on inferred images with Grad-CAM heatmap and Bounding Box in the affected area.

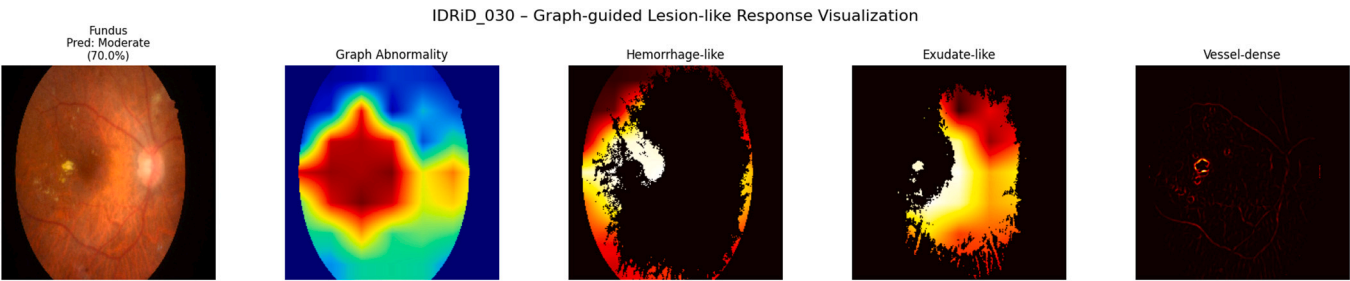


Fig. 24. Grad-CAM Abnormality Maps Across Severity Levels.

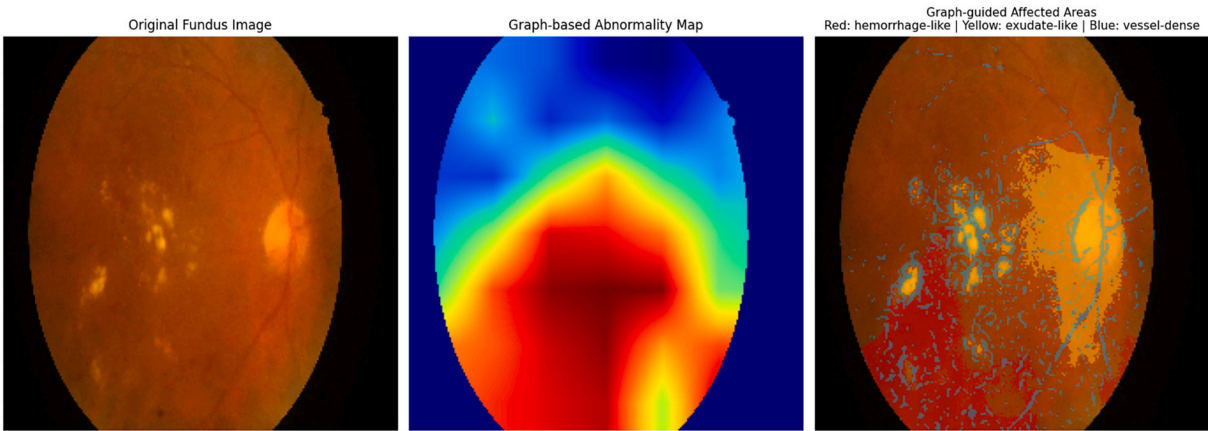


Fig. 25. Graph-Guided Abnormality Localization Using ResNet-GAT.

The figure illustrates that ResNet-GAT can detect spatially related patterns of pathology, as opposed to individual activations. The behavior is especially relevant to intermediate stages of DR, where the severity is determined by the co-occurrence of lesions, as well as spatial organization. The visualization based on graphs gives substantial qualitative data on the effectiveness of relational reasoning to enhance the stability of classification.

All visualizations mentioned above show that the proposed framework, in addition to having a high quantitative performance, is also making decisions based on clinically meaningful retinal structures. These findings establish clear and credible evidence that the model predictions are in line with already known clinical knowledge of

diabetic retinopathy by quantified lesion-aligned attention, severity-conscious progression, and relational lesion reasoning. This interpretability is essential to the successful clinical implementation of automated DR screening systems.

The three core contributions of this study are validated by the results shown in Figs. 4–7 and Tables 3–5. First, the comparative analysis of ResNet-50, Swin Transformer, and ResNet-GAT shows that local lesion extraction, global contextual modeling, and relational reasoning touch on different aspects of DR pathology, with the hybrid architecture obtaining the best multiclass performance and the greatest QWK consistency.

Second, the abnormality maps guided by graphs and the ResNet-GAT

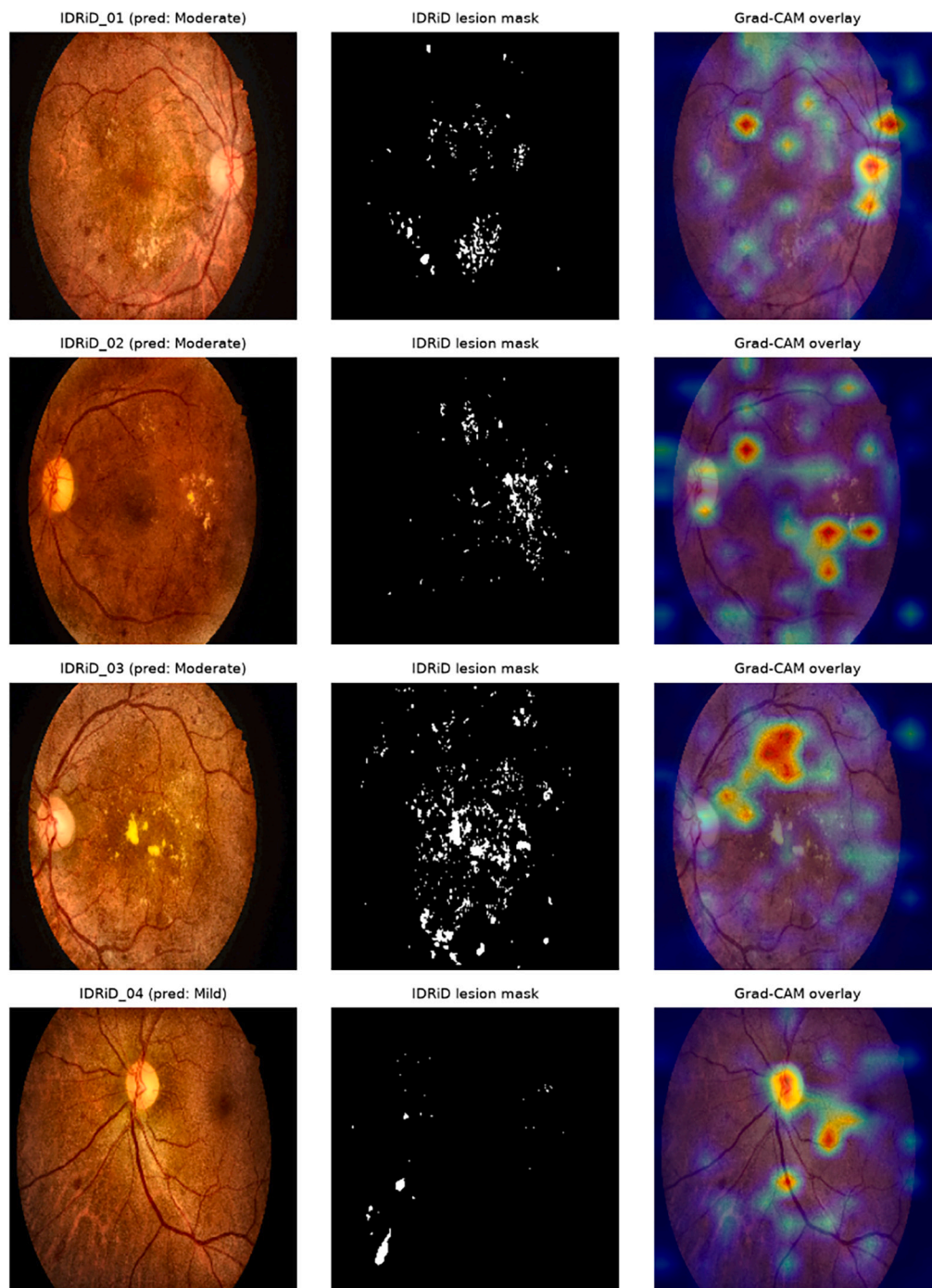


Fig. 26. Model attention vs. expert annotations. Example IDRI cases (rows) showing, left to right, the fundus image, ground-truth lesion mask, and attention heat-map.

results (Fig. 23) show that within-lesion inter-relations modeling is helpful for stability, where clinical difficulties arise in the Mild and Moderate stages of DR.

Finally, the Grad-CAM visualizations (Figs. 23–24) show, within the lesions, and focally and progressively with the severity of DR, that to improve clinically aligned, interpretable progressive lesion-targeted patterns, the additional quantitative performance to the DR classifier is gained in realism.

5.7. Quantitative explainability evaluation

To ensure that the network is focused on true pathology and not on general texture or illumination, the model's attention was quantitatively compared to the pixel-wise lesion labels of the IDRiD segmentation set of 81 fundus images with expert labels for microaneurysms, hemorrhages, hard exudates, and soft exudates. Following Quéllec et al. (Quéllec et al., 2021), Vasireddi et al. (Vasireddi et al., 2024), and lesion-grounded methodology of Herrero-Tudela (b) et al. (Herrero-Tudela et al., 2025), three complementary measures were calculated on the union of the lesion masks: (i) pointing-game accuracy, which was the percentage of images for which the peak attention was inside a lesion, using a 15-pixel tolerance, which reflects the resolution of the attention map; (ii) fraction of total attention mass within lesions compared to a chance value, based on the lesion-area fraction; and (iii) a class-conditional statistical analysis of the per-lesion attention distribution, stratified by predicted severity. The attention maps were computed using LayerCAM on the convolutional (ResNet) branch, with the same preprocessing as during training to align the maps and masks spatially.

MA = microaneurysms, HE = hemorrhages, EX = hard exudates, SE = soft exudates. Note the HE/EX rise with severity; MA stays flat.

Fig. 26 represents the example-by-example visual evidence of where the model looks vs. actual lesion locations.

Fig. 27 represents the quantitative summary; it aggregates how much attention mass falls on each lesion type across severity classes.

From Table 11, the model's attention is directed towards a lesion in the majority of images (53.1%), and the attention-in-lesion mass (2.6%) is larger than the random mass (2.2%), suggesting that the model is preferentially, but not exclusively, focused on pathological areas. Analytically more informative, the class-conditional analysis (Table 12, Fig. 27) shows a clear tendency to increase the attention with severity to hemorrhages, from 0.2% of the attention mass in cases predicted as Mild to 3.6% in Severe and Proliferative cases, and to increase attention on hard exudates from 0.3% to 1.3–1.4% over the same range. The model therefore shifts more of its diagnostic attention to the larger, more

Table 11

Attention-lesion Alignment (IDRiD Segmentation, n = 81).

Measure	Value
Pointing-game accuracy (≤ 15 px tolerance)	53.1%
Attention-in-lesion (mass fraction)	2.6%
Chance baseline (mean lesion-area fraction)	2.2%
Images evaluated	81

Table 12

Class-conditional Attention (mean % of attention mass per lesion type, by predicted grade).

Predicted Grade	n	MA	HE	EX	SE
No DR	4	0.1	0.4	0.1	0.5
Mild	17	0.1	0.2	0.3	0.4
Moderate	47	0.1	0.9	1.4	0.4
Severe	6	0.2	3.6	0.9	0.3
Proliferative	7	0.2	3.6	1.3	0.6

advanced lesions that typify higher grades of DR (representative overlays in Fig. 26).

The following two restrictions temper the results. Microaneurysms (which are just a few pixels in size) are not localized, meaning that their attention share is close to zero across grades, and are therefore not represented in the quantitative lesion evidence in the same way as hemorrhages and exudates are. Second, the analysis is based on the ResNet branch only, and some attention is placed on the optic disc, which is a normal salient landmark, meaning that the IoU is low, which is typical for small lesions with attention at the region level. Despite these limitations, the localisation performed using the tolerance parameter and the uniform class-conditional shift offers quantitative support that goes beyond the qualitative heatmaps to suggest that the model's decisions are based on structures related to the disease.

Despite the success of the proposed model, a few limitations of this study should be noted. To begin with, all the experiments for APTOS were performed using a train–test split with stratified two-fold cross-validation. While the preservation of the class distribution within each fold, and the reuse of the same folds across all designs, may be beneficial in interpreting the results, a two-fold, design is less likely to yield a fair comparison as it assigns only half of the data to training and is more sensitive to the specific partition of the data than the standard five- or ten-fold designs used in medical imaging studies. All models, however, were trained on a Tesla T4 GPU with the fixed runtime limitations of Kaggle, and this study has not been able to re-run the full multi-

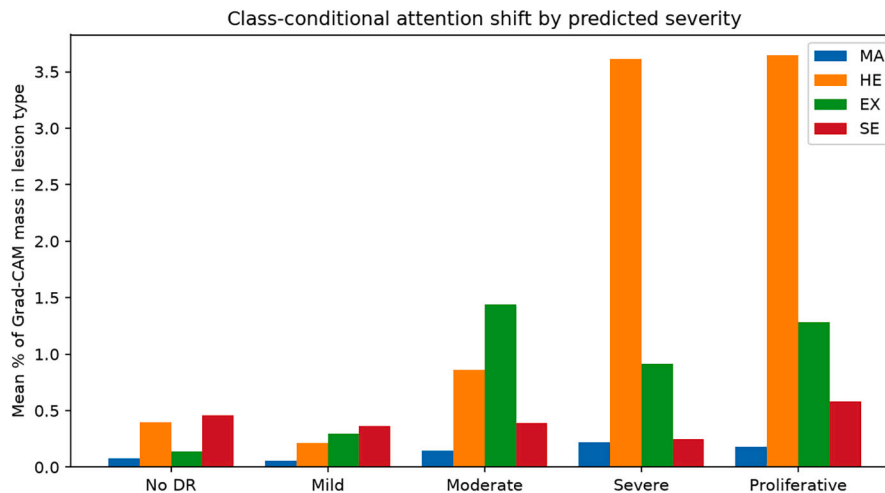


Fig. 27. Class-conditional attention shift. Mean % of LayerCAM (layer-3) saliency mass on each IDRiD lesion type (MA, HE, EX, SE) by predicted DR grade. Attention on hemorrhages and exudates rises with severity (HE: 0.4% → 3.6%), confirming the model relies on clinically relevant lesions rather than global texture.

paradigm pipeline with a five-fold or ten-fold larger computed budget. The experiment has this clearly stated in the limitations and identified access to larger computing resources as a concrete next step with confidence, for a full five-fold evaluation with confidence intervals and significance testing.

In general, the results indicate that the optimal combination of local lesion sensitivity, global contextual awareness, and relational reasoning is required to achieve successful and reliable DR severity grading, which is a solid basis for clinically implementable automated screening systems.

6. Conclusion

The present paper has included a systematically designed and understandable framework of automated diabetic retinopathy (DR) detection and severity grading through a systematic combination of convolutional, transformer-based, and graph-based learning frameworks. The proposed hybrid ResNet-Swin architecture achieves positive results in classification performance and ordinal agreement, illustrated by high Quadratic Weighted Kappa scores, confirming its clinically relevant severity progression. Similarly, the ResNet-GAT framework reinforced the importance of relational lesion modeling, particularly for stability improvement in the intermediate DR stages. Additionally, performance increases due to clinically relevant feature attribution were demonstrated by Grad-CAM and graph-guided visualizations, further enhancing transparency and the promising potential for real-world implementations. This study established a comprehensive framework that integrates local, global, and relational representation, moving towards solid, interpretable, and clinically meaningful automated DR screening systems. The framework attains real-time inference (34 ms/image) on a single GPU, and quantization and pruning are identified as future directions for edge deployment in low-resource DR screening.

Computational optimization for clinical integration in screening contexts with limited resources, improved relational modeling through adaptive graph construction, and extensive multi-institutional testing aimed at better generalization across various imaging conditions will be the primary focus for future research. Future evaluations will adopt a five-fold or ten-fold stratified cross-validation and report every metric for better stability and trust. Semi-supervised learning will likely enhance scalability and clinical relevance by applying the framework to multimodal retinal data.

CRedit authorship contribution statement

Muradul Islam: Writing – original draft, Software, Methodology, Conceptualization. **Marisa Parvez Oishy:** Software, Methodology, Formal analysis, Conceptualization. **Ferdous Anam Jibon:** Visualization, Supervision, Methodology, Data curation. **Utpal Kanti Das:** Writing – review & editing, Supervision, Investigation. **A.H.M. Kamal:** Writing – review & editing, Project administration, Funding acquisition. **Mohammad Mehedi Hassan:** Writing – original draft, Software, Project administration, Formal analysis, Conceptualization. **Gahangir Hossain:** Writing – review & editing, Validation, Investigation.

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Declaration of competing interest

The authors declare that they have no competing financial or personal interests. The authors are engaged in related research activities; however, these activities did not influence the design, execution, analysis, or interpretation of the current study.

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Data availability

We used the public datasets: APTOS (Maggie and Sohier, 2019) and Indian Diabetic Retinopathy Image Dataset (IDRiD) (Prasanna et al., 2018), which are available online.

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